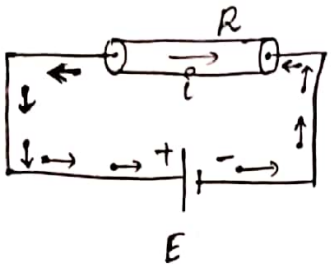
Alternating Current (A·C)

30|06|25| Monday

*



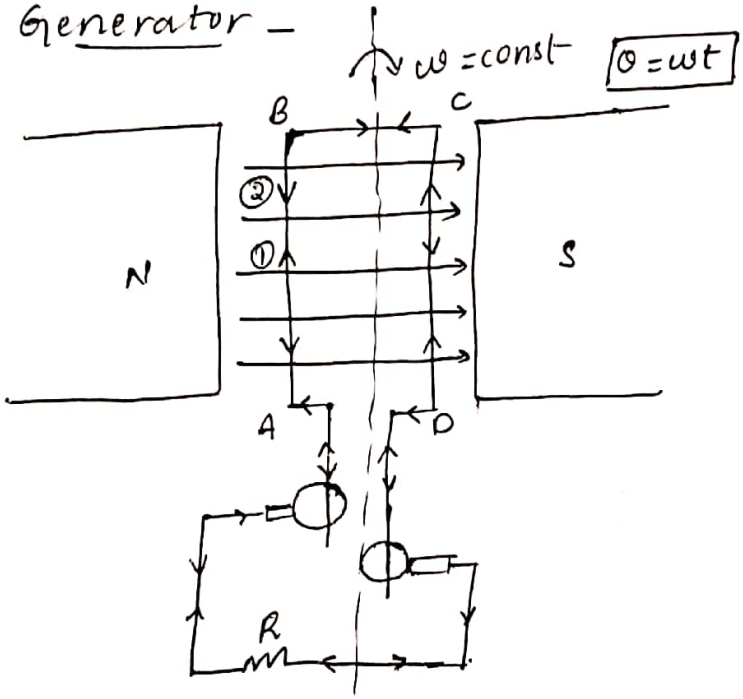
$i = \frac{E}{R} = \text{const}$ (d·c)

* varying currents -

eg: (1) $i = 2t$

(2) $i = i_0(1 - e^{-\frac{R}{L}\cdot t})$ } Here magnitude is changing but direction is same.

* A·C Generator -



* Direction of induced emf / induced current is given by - Fleming's right hand rule.

fore finger direction — magnetic field

thumb direction — motion of conductor

middle finger direction — direction of emf / induced current.

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* At any time t,

θ is angle b/w $\bar{A}$ & $\bar{B}$.

∴ flux linked with the coil is,

$$\phi = N(\bar{A}.\bar{B})$$

$$\phi = NAB\cos\theta$$

$$\boxed{\phi = NAB\cos(\omega t)}$$

$\phi = NAB$ = max flux

* Induced emf in coil is,

$$e = \frac{-d\phi}{dt}$$

$$= -\frac{d}{dt}(NAB\cos\omega t)$$

$$= -NAB(-\sin\omega t)\,\omega$$

$$e = NA\omega B\sin\omega t \Rightarrow \boxed{e = e_0\sin\omega t}$$

$\boxed{e_0 = NA\omega B}$ = peak / max value of emf.

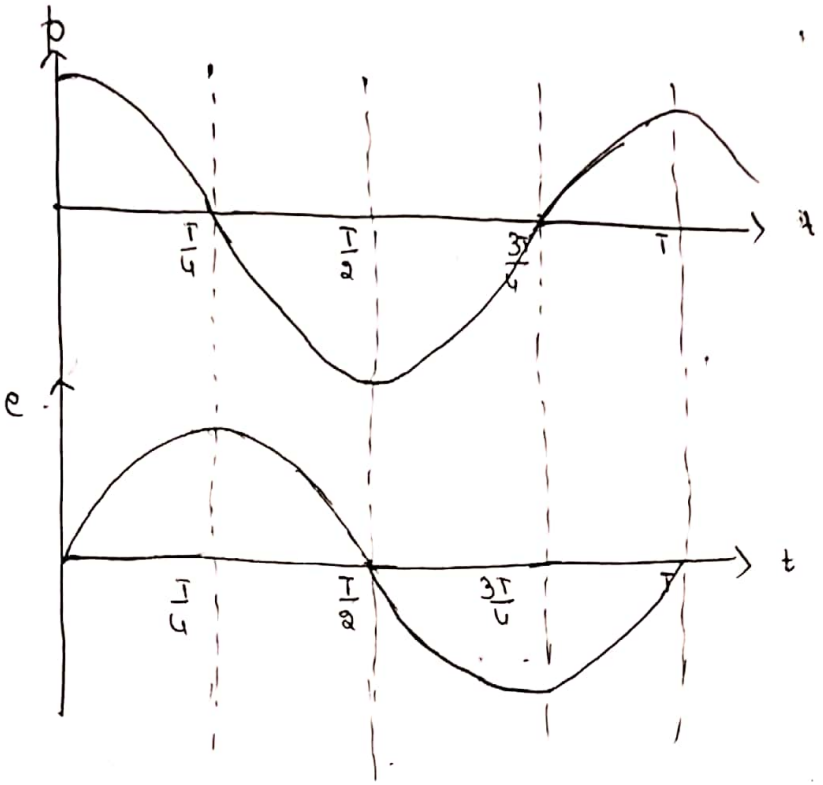
* Induced current,

$$i = \frac{e}{R} = \frac{e_0}{R}\sin\omega t = i_0\sin\omega t$$

$\boxed{i_0 = \frac{e_0}{R}}$ = peak value of current.

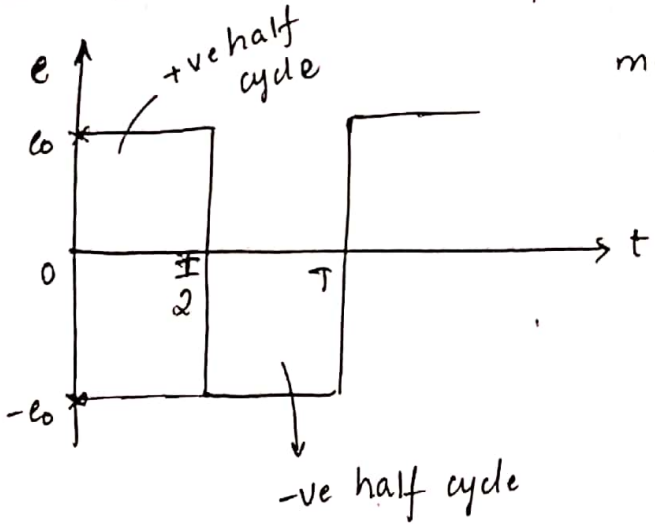
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* Graph :



Another forms of A.C :

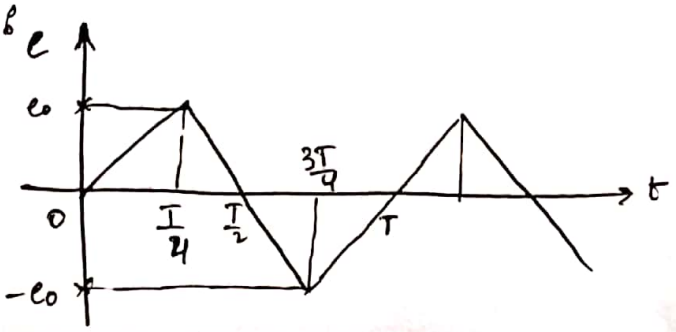
1) square wave :



magnitude - same

direction - change

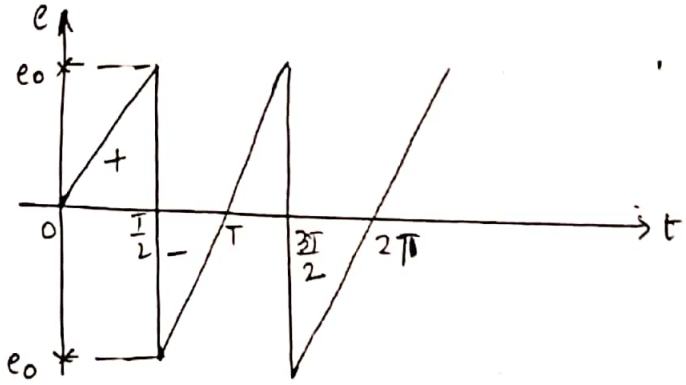
2) Triangular wave :



magnitude & direction both - change

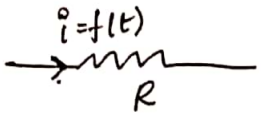
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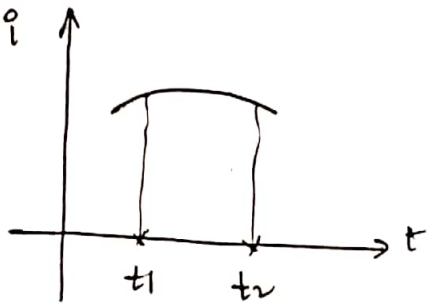
3) Saw-Tooth Wave :



Average Current / Mean current –

Let $i = f(t)$





* charge flowing through resistance R b|w $t = t_1$ & $t = t_2$ is

$$\Delta q_1 = \int_{t_1}^{t_2} i \cdot dt = \text{area of } i\text{-}t \text{ graph b|w } t = t_1 \text{ \& } t = t_2$$

* Charge flowing through same resistance R in the same duration due to D.C is –

$$\Delta q_2 = \int_{t_1}^{t_2} i_{dc} \cdot dt = i_{dc}(t_2 - t_1) \quad -②$$

* If $\Delta q_1 = \Delta q_2$ then $\boxed{i_{dc} = i_{avg}}$

$$\therefore i_{dc} \times (t_2 - t_1) = \int i_{dc} \cdot dt$$

$$i_{avg}(t_2 - t_1) = \int_{t_1}^{t_2} i \cdot dt$$

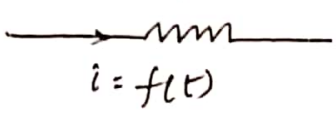
$$i_{avg} = \frac{\int_{t_1}^{t_2} i \cdot dt}{\int_{t_1}^{t_2} dt} = \frac{\int_{t_1}^{t_2} i \cdot dt}{t_2 - t_1} = \frac{\text{Area of } i\text{-}t \text{ graph}}{t_2 - t_1}$$

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★ RMS current :-

└ also called effective / virtual current.

Let $i = f(t)$



★ Heat produced in resistance 'R' blw $t=t_1$ & $t=t_2$ is

$$H_1 = \int_{t_1}^{t_2} i^2 R\,dt \quad -①$$

Heat produced in the same resistance 'R' & in the same duration due to dc is,

$$H_2 = \int_{t_1}^{t_2} i_{dc}^2 R \cdot dt$$

$$H_2 = i_{dc}^2 R(t_2 - t_1) \quad -②$$

If $H_1 = H_2$ then $\boxed{i_{dc} = i_{RMS}}$

$$i_{dc}^2 (R)(t_2 - t_1) = \int_{t_1}^{t_2} i^2 R \cdot dt$$

$$i_{rms}^2 = \frac{\int_{t_1}^{t_2} i^2 \cdot dt}{t_2 - t_1}$$

$$\boxed{i_{rms} = \sqrt{\frac{\text{area of } i\text{-}t \text{ graph}}{t_2 - t_1}}}$$

Ex:

1) $i = 2t$. Find ① Avg current blw $t=1s$ & $t=3sec$

② Rms current blw $t=1sec$ & $t=3sec$

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Sol⁸

1) M-I : Analytical method :

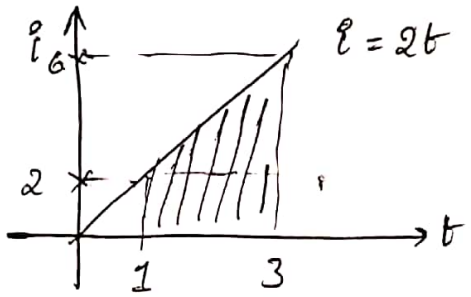
$$i_{avg} = \frac{\int_1^3 i\,dt}{\int_1^3 dt}$$

$$= \frac{\int_1^3 2t}{\int_1^3 .dt}$$

$$= \frac{\frac{2}{2}(9-1)}{2}$$

$$= \frac{8}{2} = 4A$$

M-II : graphical method :



$$i_{avg} = \frac{\frac{1}{2} \times 2 \times 4 + 2 \times 2}{2}$$

$$= \frac{8}{2} = 4$$

2) M-I :

$$i_{rms} = \sqrt{\frac{\int_1^3 i^2 .dt}{\int_1^3 dt}} = \sqrt{\frac{4\int_1^3 t^2 .dt}{2}}$$

$$= \sqrt{\frac{2}{3}(27-1)}$$

$$= \sqrt{\frac{26 \times 2}{3}} = 2\sqrt{13/3}\ A$$

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M-II 6

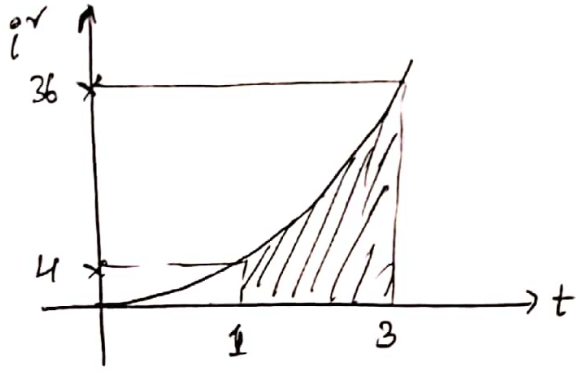
$i = 2t$

$i^2 = 4t^2$

for $y = kx^2$

$$\text{Area} = \int_0^x y \cdot dx$$

$$= \frac{k x^3}{3}$$



Area = $\frac{1}{3}$ x base x height

here

$$i_{rms} = \sqrt{\frac{\frac{1}{3}\times 3\times 36 - \frac{1}{3}\times 1\times 4}{2}}$$

$$= \sqrt{\frac{4^2}{3}\frac{(27-1)}{2}}$$

$$= \frac{2\sqrt{13}}{\sqrt{3}}.$$

Ex: For $i = 2+3t$, Find Avg value and Rms value b/w $t = 1$ sec & $t = 3$ sec.

Sol: M-II 6

$$i_{avg} = \frac{\int_1^3 i \cdot dt}{\int_1^3 dt}$$

$$= \frac{\int_1^3 (2+3t) \cdot dt}{2}$$

$$= \frac{2(2) + \frac{3}{2}(9-1)}{2}$$

$$= \frac{4 + 3(4)}{2} = 8.$$

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$$i_{rms} = \sqrt{\frac{\int_1^3 i^2 \cdot dt}{\int_1^3 dt}}$$

$$= \sqrt{\frac{\int_1^3 (2+3t)^2 \cdot dt}{2}}$$

$$= \sqrt{\frac{\left[(2+3t)^3\right]_1^3 \cdot \frac{1}{3} \times \frac{1}{3}}{2}}$$

$$= \sqrt{\frac{(2+9)^3 - (2+3)^3}{18}}$$

$$= \sqrt{\frac{11 \times 11 \times 11 - 5 \times 5 \times 5}{18}}$$

$$= \sqrt{\frac{121 \times 11 - 125}{18}}$$

$$= \sqrt{\frac{121 \times 11 - 121 - 4}{18}}$$

$$= \sqrt{\frac{121\,(10) - 4}{18}}$$

$$= \sqrt{\frac{121 \times 5 - 2}{9}}$$

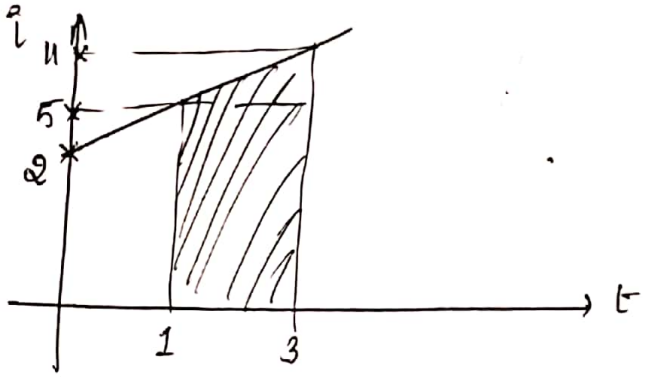
$$= \sqrt{\frac{605 - 2}{9}}$$

$$= \sqrt{\frac{603}{9}} = \sqrt{67}$$

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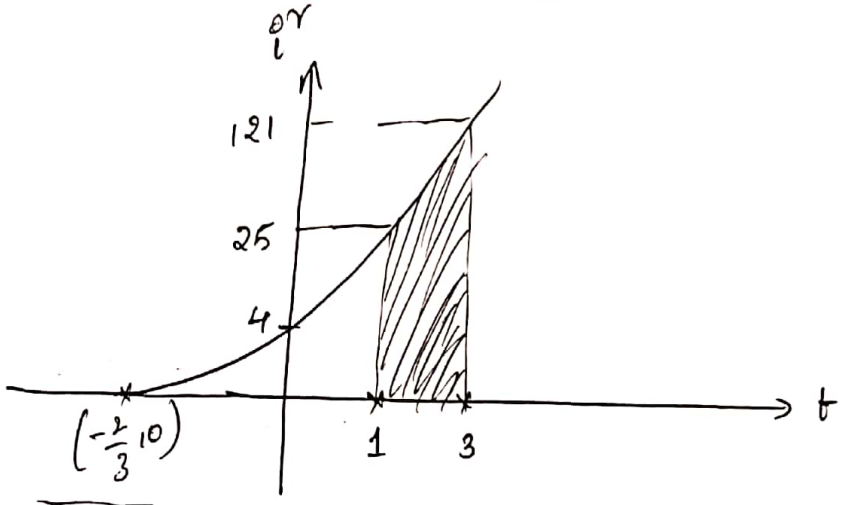
M-II

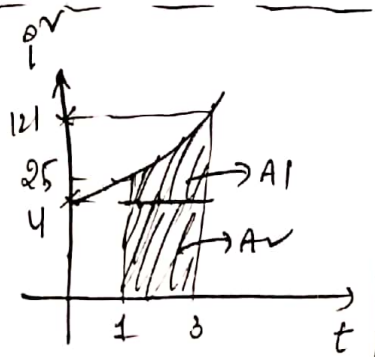
Graphical Method:



$$i_{avg} = \frac{\text{Area}}{3-1} = \frac{\frac{1}{2}\times 2\times 6 + 5\times 2}{2}$$

$$= \frac{16}{2} = 8$$





$A_1 = \frac{1}{3}\times 3(121-4) - \frac{1}{3}(1)(25-4)$

$A_2 = 4\times 2$

$A = A_1 + A_2$

$i_{rms} = \sqrt{\frac{A}{2}} = \sqrt{59}$

wrong

$$i_{rms} = \sqrt{\frac{\frac{1}{3}\times\left(3+\frac{2}{3}\right)(121) - \frac{1}{3}\left(1+\frac{2}{3}\right)(25)}{3-1}}$$

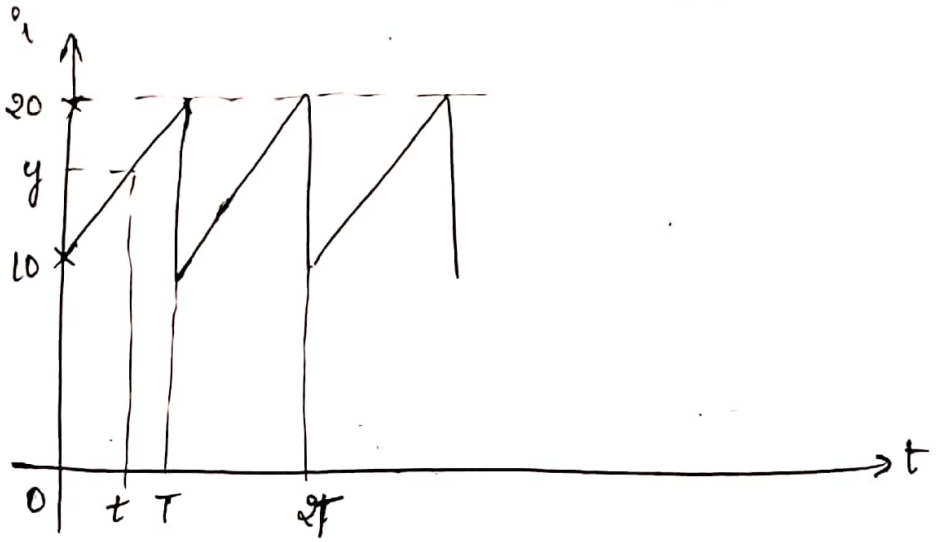
$$= \sqrt{\frac{11\times 121 - 5\times 25}{3\times 3\times 2}}$$

$$= \sqrt{\frac{603}{9}}$$

$$= \sqrt{67}$$

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Ex 6



Sol 6 M-2 eqn of wave form,

$$y = mt + c$$

$$y = \frac{(20-10)}{T-0}\,t + 10$$

$$y = \frac{10}{T}\,t + 10$$

$$i_{avg} = \frac{\int_0^T i\,dt}{\int_0^T dt} = \frac{\int_0^T \left(\frac{10}{T}t + 10\right).dt}{T}$$

$$= \frac{\frac{10}{T}\left(\frac{T^2}{2}\right) + 10T}{T}$$

$$= 5 + 10 = 15$$

$$i_{rms} = \sqrt{\frac{\int_0^T i^2.dt}{\int_0^T .dt}} = 15\sqrt{\frac{\int_0^T \left(\frac{10}{T}t + 10\right)^2.dt}{T}}$$

$$= \sqrt{\frac{\left[\left(\frac{10}{T}t + 10\right)^3\right]_0^T \times \frac{T}{10} \times \frac{1}{3}}{T}}$$

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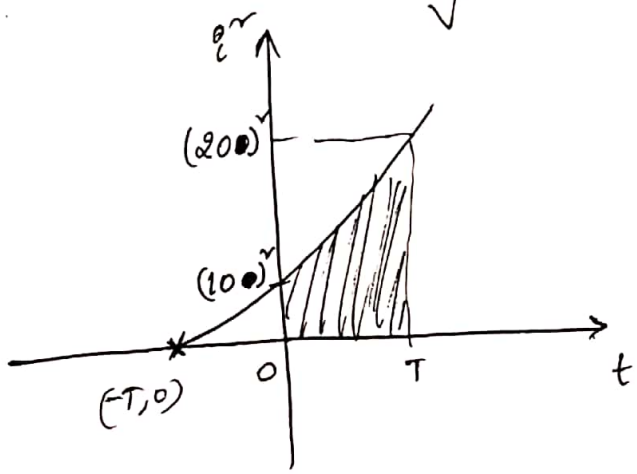
$$i_{rms} = \sqrt{\frac{800-100}{3}} = 10\sqrt{\frac{7}{3}}.$$

M-II :

$$i_{avg} = \frac{\text{Area of graph}}{T} = \frac{\frac{1}{2}\times T\times 10 + (T)(10)}{T}$$

$$= 5+10 = 15.$$

$$i_{rms} = \sqrt{\frac{\text{Area of graph}}{T}}$$



$$\text{Area} = \frac{1}{3}(2T)(200)^2 - \frac{1}{3}(T)(100)^2$$

$$= \frac{1}{3}T\,(80000 - 10000)$$

$$= \frac{700}{3}T$$

$$i_{rms} = \sqrt{\frac{700\cancel{T}}{3\cancel{T}}}$$

$$= \frac{10}{\sqrt{3}}\sqrt{7}$$

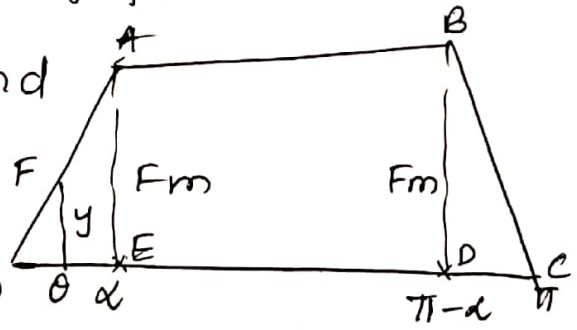
$$\boxed{i_{rms} = 10\sqrt{\frac{7}{3}}}$$

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Ex 6 The half cycle of an alternating signal is as shown. It increases linearly from zero at 0° to Fm at $\alpha°$. It remains const from $\alpha°$ to $(180-\alpha)°$ and decreasing uniformly from Fm to zero at 180°. Find

1) effective value of signal.

2) Avg value of signal.



Sol: M-I

$$\frac{y}{\theta} = \frac{Fm}{\alpha}$$

$$y = \frac{Fm}{\alpha}(\theta)$$

$$y_1 = \frac{Fm\,\theta}{\alpha} \qquad 0 \le \theta \le \alpha$$

$$y_2 = Fm \qquad \alpha \le \theta \le \pi - \alpha$$

$$y_3 = \frac{Fm\,\theta}{\alpha} \qquad \pi - \alpha \le \theta \le \pi$$

∴ Avg value,

$$y = \frac{\int_0^{\alpha} \frac{Fm\,\theta}{\alpha}\cdot d\theta + \int_{\alpha}^{\pi-\alpha} Fm\cdot d\theta + \int_{\pi-\alpha}^{\pi} \frac{Fm\,\theta}{\alpha}\cdot d\theta}{\pi}$$

$$= \frac{\frac{Fm}{\alpha}\left(\frac{\alpha^2}{2}\right) + Fm(\pi - 2\alpha) + \frac{Fm}{\alpha}\left(\frac{\pi^2 - (\pi-\alpha)^2}{2}\right)}{\pi}$$

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$$\times \left(= \frac{\frac{F_m \alpha}{2} + F_m(\pi - 2\alpha) + \frac{F_m}{2\alpha}(2\pi - \alpha)(+\alpha)}{\pi} \right.$$

$$= \left. \frac{\frac{F_m \alpha}{2} + F_m \pi - 2\alpha F_m + \frac{F_m \alpha}{2}}{\pi} \right) \times$$

$$= \frac{2 F_m \pi - 2\alpha F_m}{\pi}$$

$$\boxed{I_{avg} = \frac{2 F_m (\pi - \alpha)}{\pi}}$$

$$I_{rms} = \sqrt{\frac{2\int_0^{\alpha} y_1^2 \cdot d\theta + \int_{\alpha}^{\pi-\alpha} y_2^2 \cdot d\theta}{\pi}}$$

$$= \sqrt{\frac{2\frac{F_m^2}{\alpha^2}\int_0^{\alpha} \theta^2 \cdot d\theta + F_m^2 \int_{\alpha}^{\pi-\alpha} d\theta}{\pi}}$$

$$= \sqrt{\frac{2\frac{F_m^2}{\alpha^2}\frac{\alpha^3}{3} + F_m^2(\pi - 2\alpha)}{\pi}}$$

$$= \sqrt{\frac{F_m^2\left(\frac{2\alpha}{3} + \pi - 2\alpha\right)}{\pi}}$$

$$\boxed{I_{rms} = F_m \sqrt{\frac{3\pi - 4\alpha}{3\pi}}}$$

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M-II 6

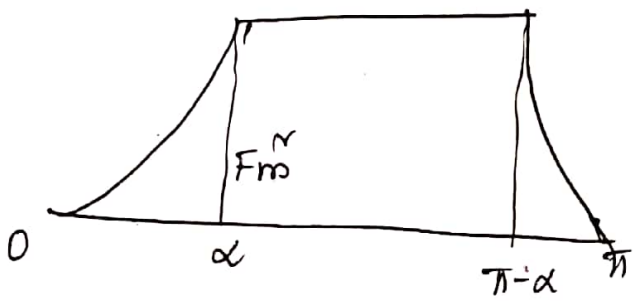
1) $i_{avg} = \dfrac{\text{area of trapezium}}{\pi}$

$$= \frac{\frac{1}{2} \times \alpha \times Fm + (\pi - 2\alpha) Fm + \frac{1}{2} \times \alpha (Fm)}{\pi}$$

$$= \frac{\alpha Fm + \pi Fm - 2\alpha Fm}{\pi}$$

$$= \frac{Fm(\pi - \alpha)}{\pi}$$

2)



$$i_{rms} = \sqrt{\frac{\frac{1}{3}\alpha Fm^2 + Fm^2(\pi - 2\alpha) + \frac{1}{3}\alpha Fm^2}{\pi}}$$

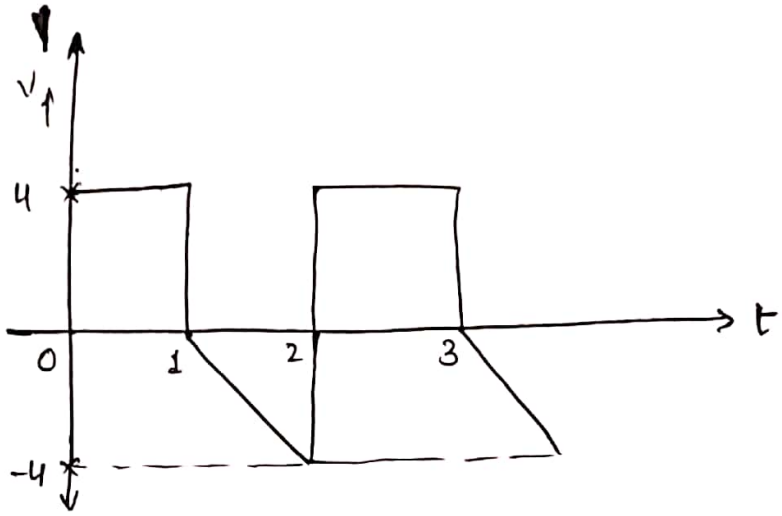
$$= Fm \sqrt{\frac{\frac{\alpha}{3} + \pi - 2\alpha + \frac{1}{3}\alpha}{\pi}}$$

$$= Fm \sqrt{\frac{\pi - \frac{4\alpha}{3}}{\pi}}$$

$$= Fm \sqrt{\frac{3\pi - 4\alpha}{3\pi}}$$

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Ex:



Find

1) Avg value

2) rms value of given cycle.

Sol:

$V_1 = 4 \qquad 0 \le t \le 1$

$V_2 = mt + c \qquad 1 \le t \le 2$

$0 = m(1) + c$

$-4 = m(2) + c$

$c = 4$

$m = -4$

$\Big\} \; V_2 = -4t + 4$

1) Avg value,

$$V_{avg} = \frac{\int_0^2 v \cdot dt}{\int_0^2 dt} = \frac{\int_0^1 V_1 \cdot dt + \int_1^2 V_2 \cdot dt}{(2-0)}$$

$$V_{avg} = \frac{4 + [-2t^2 + 4t]_1^2}{2}$$

$$= \frac{+4 + -2(4-1) + 4(1)}{2}$$

$$= \frac{8-6}{2} = \frac{2}{2} = 1$$

2) rms value,

$$V_{rms} = \sqrt{\frac{\int_0^1 V_1^2 \cdot dt + \int_1^2 V_2^2 \cdot dt}{2-0}}$$

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$$= \sqrt{\frac{\int_0^1 4^2 \cdot dt + \int_1^2 (-4t+4)^2 \cdot dt}{2}}$$

$$= \sqrt{\frac{16(1) + \left[(-4t+4)^3\right]_1^2 \times \frac{1}{-12}}{2}}$$

$$= \sqrt{\frac{16 + \left(\frac{1}{12}\right)(16) + 0}{2}}$$

$$= \sqrt{\frac{24 \times 16}{2 \times 3}} = \frac{8}{\sqrt{6}} = 4\sqrt{\frac{2}{3}}$$

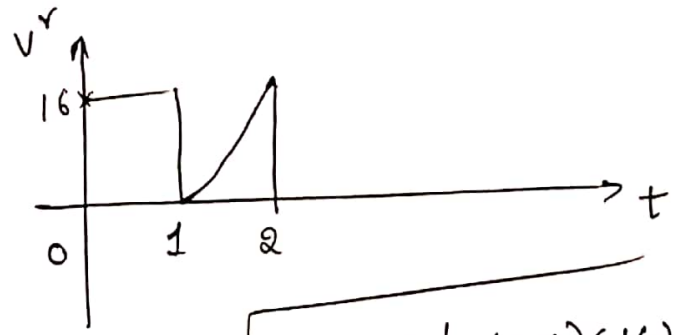
M-II 6

Graphical Method :

1) $V_{avg} = \frac{\text{Area}}{2-0} = \frac{4 \times 1 + \frac{1}{2} \times 1 \times (-4)^2}{2}$

$= 1\ A$

2)

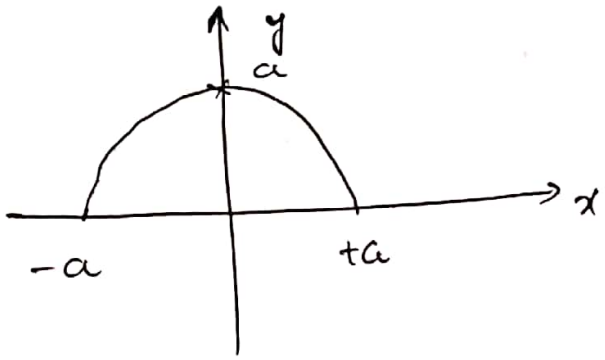


$$V_{rms} = \sqrt{\frac{16 \times 1 + \frac{1}{3}(2-1)(16)}{2}}$$

$$= \sqrt{\frac{16 + \frac{16}{3}}{2}} = 4\sqrt{\frac{2}{3}}$$

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Ex 6 Find the rms value of current whose wave form is ~~same~~ semicircle with amplitude 'a'.



Sol^{n} $x^2 + y^2 = a^2$

$y^2 = a^2 - x^2$

now, $I_{rms} = \sqrt{\dfrac{\int_{x=-a}^{x=+a} y^2 \, dx}{\int_{x=-a}^{x=+a} dx}}$

$= \sqrt{\dfrac{\int_{x=-a}^{a} (a^2 - x^2) \, dx}{2a}}$

$= \sqrt{\dfrac{\left[a^2 x - \dfrac{x^3}{3}\right]_{-a}^{a}}{2a}}$

$= \sqrt{\dfrac{a^2(2a) - \dfrac{a^3 - (-a)^3}{3}}{2a}}$

$= \sqrt{\dfrac{2a^3 - \dfrac{2a^3}{3}}{2a}} = \sqrt{\dfrac{2a^2}{3}} = a\sqrt{\dfrac{2}{3}}$

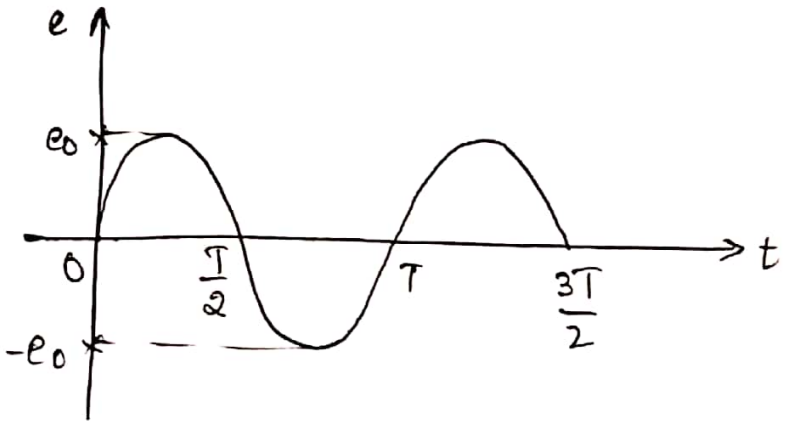
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We have, eqn of alternating emf is $e = e_0 \sin \omega t$

$e_0 \longrightarrow$ peak value

$\omega = 2\pi f = \frac{2\pi}{T}$ = angular frequency

ωt = phase (rad)



→ Avg value of emf during +ve half cycle

$$e_{avg} = \frac{\int_0^{T/2} e\,dt}{\int_0^{T/2} dt}$$

$$= \frac{e_0 \int_0^{T/2} \sin \omega t \cdot dt}{\frac{T}{2}}$$

$$= \frac{2e_0}{T} \frac{[-\cos \omega t]_0^{T/2}}{\omega}$$

$$= \frac{2e_0}{2\pi} \left[-\cos \omega \frac{T}{2} + \cos 0\right]$$

$$= \frac{e_0}{\pi} (1+1) = \frac{2e_0}{\pi}$$

$$\boxed{e_{avg} = \frac{2e_0}{\pi}}$$

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$$\frac{2e_0}{\pi} = \frac{\text{area}}{T/2}$$

$$\boxed{\frac{e_0 T}{\pi} = \text{area}}$$

Similarly, during -ve half cycle $e_{avg} = -\frac{2e_0}{\pi}$

★ for a complete cycle, $e_{avg} = 0$.

→ Rms value of emf during +ve half cycle

$$e_{rms} = \sqrt{\frac{\int_0^{T/2} e^2 \cdot dt}{\int_0^{T/2} dt}}$$

$$= \sqrt{\frac{e_0^2 \int_0^{T/2} \sin^2 \omega t \cdot dt}{T/2}}$$

$$= \sqrt{\frac{2e_0^2}{T} \int_0^{T/2} \frac{1-\cos 2\omega t}{2} \cdot dt}$$

$$= \sqrt{\frac{2e_0^2}{T} \left[\frac{T}{4} - \frac{[\sin 2\omega t]_0^{T/2}}{4\omega}\right]}$$

$$= \sqrt{\frac{2e_0^2}{T} \left[\frac{T}{4} - \frac{[\sin \omega T - 0]}{4\omega}\right]}$$

$$= \sqrt{\frac{e_0^2}{2}}$$

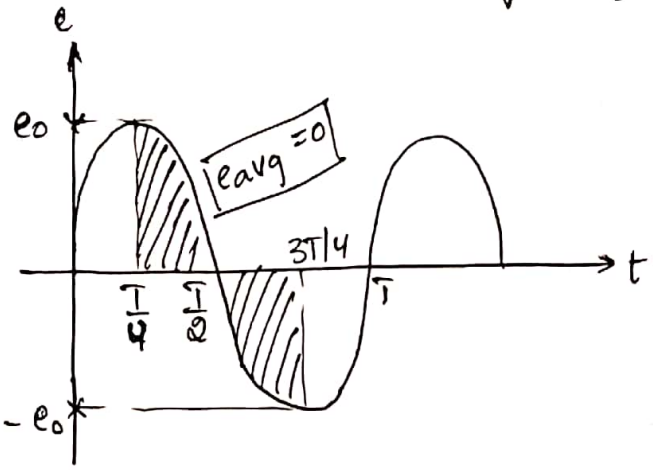
$$= \frac{e_0}{\sqrt{2}} = 0.707\, e_0$$

Similarly, during -ve half cycle $e_{rms} = +\frac{e_0}{\sqrt{2}}$

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* for a complete cycle, $e_{rms} = \frac{e_0}{\sqrt{2}}$

$$e_{rms} = \sqrt{\frac{\int_0^T e^2 . dt}{\int_0^T dt}} = \sqrt{\frac{e_0^2}{T} \times \frac{T}{2}} = \frac{e_0}{\sqrt{2}}$$



** Time taken by emf to increase from zero to maximum value, $e = e_0 \sin \omega t$

$$e_0 = e_0 \sin \omega t$$

$$\sin \omega t = 1$$

$$\omega t_1 = \frac{\pi}{2},$$

$$t_1 = \frac{\pi}{2\omega}$$

$$\boxed{t_1 = \frac{T}{4}}$$

** Time taken by emf to increase from zero to avg value, $e = e_0 \sin \omega t$

$$\frac{2e_0}{\pi} = e_0 \sin \omega t$$

$$\sin \omega t = \frac{2}{\pi}$$

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$$\boxed{t_2 = \frac{1}{\omega}\sin^{-1}\left(\frac{2}{\pi}\right)}$$

*** Time taken by emf to increase from zero to rms value,

$$e = e_0 \sin\omega t$$

$$\frac{e_0}{\sqrt{2}} = e_0 \sin\omega t$$

$$\omega t = \frac{\pi}{4}$$

$$t = \frac{\pi}{4\omega}$$

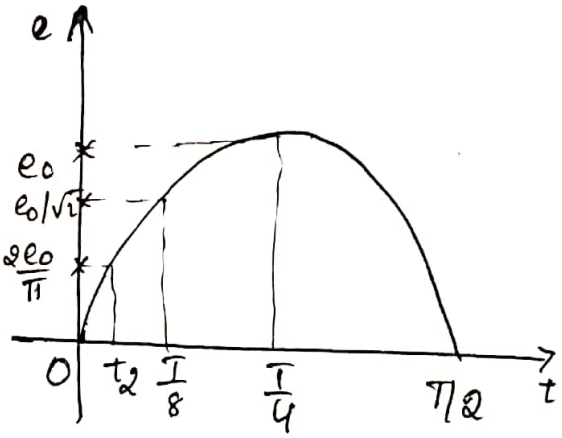
$$t_3 = \frac{\pi}{4(2\pi)} \times T$$

$$\boxed{t_3 = \frac{T}{8}}$$

clearly,

$$\boxed{t_1 > t_2 < t_3}$$

$$\Rightarrow \boxed{t_1 > t_3 > t_2}$$



*** Time taken by the emf to increase from rms value to peak value is,

$$\Delta t = t_1 - t_3$$

$$= \frac{T}{4} - \frac{T}{8}$$

$$\boxed{\Delta t = \frac{T}{8}}$$

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* Important formula :

$$\rightarrow \langle \sin\omega t \rangle = \frac{1}{T}\int_0^T \sin\omega t \cdot dt = 0$$

$$\rightarrow \langle \cos\omega t \rangle = \frac{1}{T}\int_0^T \cos\omega t \cdot dt = 0$$

$$\rightarrow \langle \sin 2\omega t \rangle = \frac{1}{T}\int_0^T \sin 2\omega t \cdot dt = 0$$

$$\rightarrow \langle \cos 2\omega t \rangle = \frac{1}{T}\int_0^T \cos 2\omega t \cdot dt = 0$$

$$\rightarrow \langle \sin^2\omega t \rangle = \frac{1}{T}\int_0^T \sin^2\omega t \cdot dt = \frac{1}{2}$$

$$\rightarrow \langle \cos^2\omega t \rangle = \frac{1}{T}\int_0^T \cos^2\omega t \cdot dt = \frac{1}{2}$$

Power in AC circuits :

In d.c circuit,

$$P = VI = I^2R = \frac{V^2}{R}$$

In a.c circuit,

- there will be some phase difference b/w emf and current. (ϕ)

$$\phi \begin{cases} +ve \\ -ve \\ zero \end{cases}$$

* we have,

$$e = e_0 \sin\omega t$$

& $$i = i_0 \sin(\omega t + \phi)$$

** instantaneous power,

$$P_{inst} = ei$$

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$= (e_o \sin\omega t)(i_o \sin(\omega t + \phi))$

$= e_o i_o \sin\omega t \sin(\omega t + \phi)$

$= e_o i_o \sin\omega t (\sin\omega t \cos\phi + \cos\omega t * \sin\phi)$

$$\boxed{P_{inst} = e_o i_o \sin^2\omega t \cos\phi + \frac{1}{2} e_o i_o \sin 2\omega t \sin\phi}$$

* Avg power, $P_{avg} = \dfrac{\int_0^T P_{inst} \cdot dt}{\int_0^T dt}$

$$= \frac{e_o i_o \int_0^T \sin^2\omega t \cos\phi \cdot dt + \frac{1}{2} e_o i_o \sin\phi \int_0^T \sin 2\omega t}{T}$$

$$= \frac{\frac{1}{2} e_o i_o \cos\phi + \frac{1}{2} e_o i_o \sin\phi (0)}{T} \times T$$

$$= \frac{e_o i_o}{2} \cos\phi$$

$$P_{avg} = \frac{e_o}{\sqrt{2}} \times \frac{i_o}{\sqrt{2}} \cos\phi \Rightarrow \boxed{P_{avg} = P_{rms} \cos\phi}$$

$$\boxed{P_{rms} = \frac{e_o i_o}{2}}$$

True power

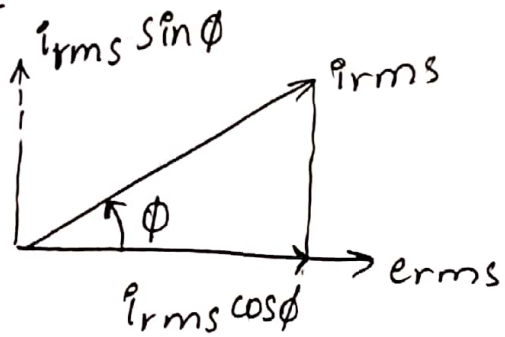
Power factor

apparent (virtual) power

* power factor has no units and dimensions.

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Wattless current :



i_{rms} is leading ϕ rad than e_{rms}.

$i_{rms}\sin\phi$ — does not consume any Power in the circuit

↓

wattless current

$i_{rms}\cos\phi$ — watt full component of current.

Phasor diagram :-

1) e_0, i_0 are considered as rotating vectors.

2) Magnitudes of e_0 & i_0 represented by length of vectors.

3) projections of e_0 & i_0 on y-axis gives "instantaneous values"

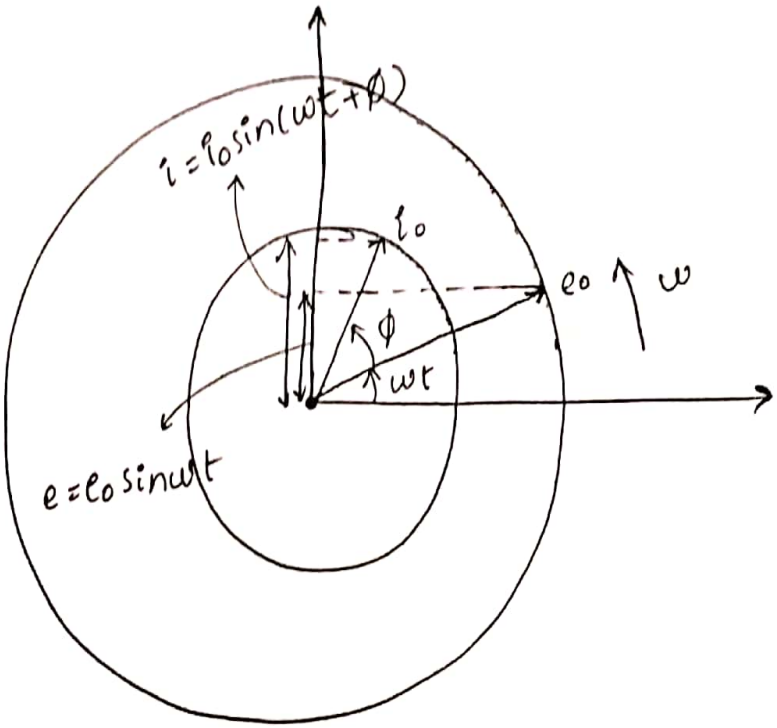
For ex :

$e = e_0 \sin \omega t$

$i = i_0 \sin(\omega t + \phi)$

$i_0 = \frac{e_0}{R} \Rightarrow i_0 < e_0$

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Points:

* Symbol for ac supply =>

* AC supply of 220V means, $V_{rms} = 220V$

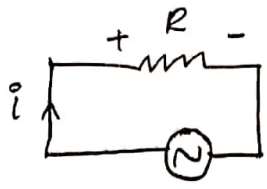
$$\frac{V_0}{\sqrt{2}} = 220$$

$$\boxed{V_0 = 220\sqrt{2}\,V}$$

1|07|25| Tuesday

AC circuit containing pure resistance (R):

$$e = e_0 \sin\omega t \quad -①$$



At any time t,

using K.V.L,

$$e - iR = 0$$

$$i = \frac{e}{R} = \frac{e_0 \sin\omega t}{R}$$

$$\boxed{i = i_0 \sin\omega t} \quad -②$$

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where $i_0 = \frac{e_0}{R}$ = peak value of current.

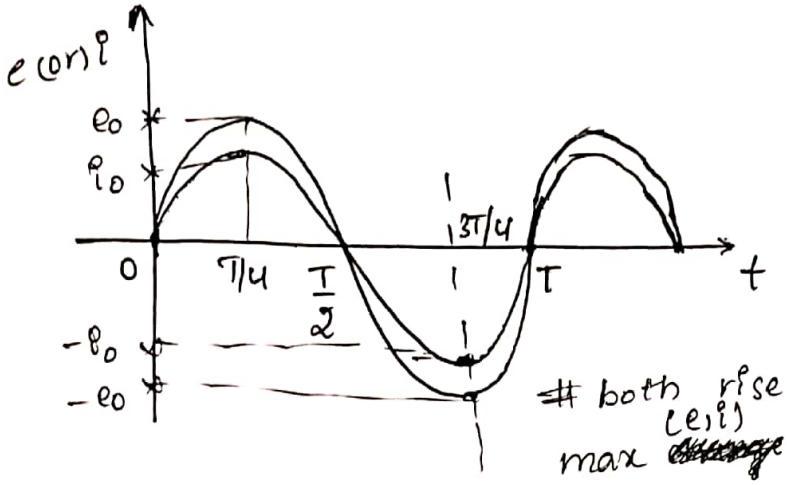
Points:

1. phase diff b/w emf and current is,

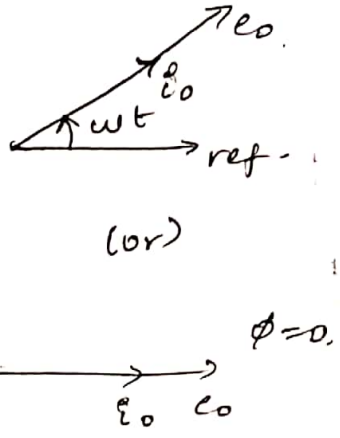
$$\boxed{\phi = 0^\circ}$$

⇒ power factor, $\cos\phi = \cos 0 = 1$

2. wave form:



Phasor diagram:



both rise to max (e,i) & drops to zero simultaneously.

3. Instantaneous power,

$$P_{inst} = ei = (e_0 \sin\omega t)(i_0 \sin\omega t)$$

$$= e_0 i_0 \sin^2\omega t$$

Average power,

$$\langle P \rangle = \frac{1}{2} e_0 i_0 \cos\phi$$

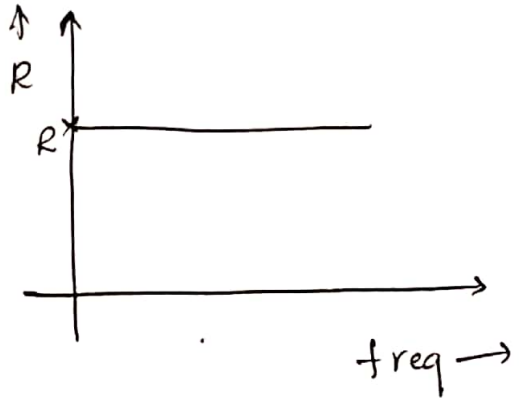
$$= \frac{1}{2} e_0 i_0 \text{ (max)}$$

* Energy produced in resistance R in time t,

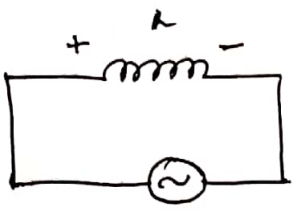
$$E = \langle P \rangle t = \frac{1}{2} e_0 i_0 t = \frac{e_0^2 t}{2R}.$$

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* Resistance 'R' doesnot depends on frequency of AC.



AC circuit containing pure inductor only :



$e = e_0 \sin\omega t$ —①

At any time t,

$$e - L\cdot\frac{di}{dt} = 0$$

$$\frac{di}{dt} = \frac{e_0}{L}\sin\omega t$$

$$\int di = \int \frac{e_0}{L}\sin\omega t \cdot dt$$

$$i = -\frac{e_0}{L\omega}\cos\omega t + c$$

To find c, we have,

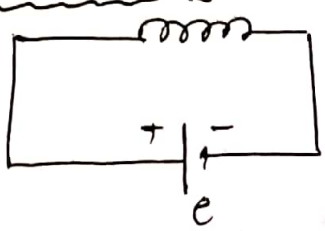
For a complete cycle,

$$\langle i \rangle = 0$$

$$\langle i \rangle = -\frac{e_0}{\omega L}\langle \cos\omega t \rangle + c$$

$$\boxed{c = 0}$$

D.C circuit:



$$E = L\cdot\frac{di}{dt}$$

$$di = \frac{E}{L}\cdot dt$$

$$\boxed{i = \frac{E}{L}\cdot t}$$

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$$i = -\frac{e_0}{\omega L}\cos\omega t$$

$$i = \frac{-e_0}{\omega L}\sin\left(\frac{\pi}{2} - \omega t\right)$$

$$i = \frac{e_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

$$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right) \quad -(2)$$

here, $i_0 = \frac{e_0}{\omega L}$ = peak value of current.

points:

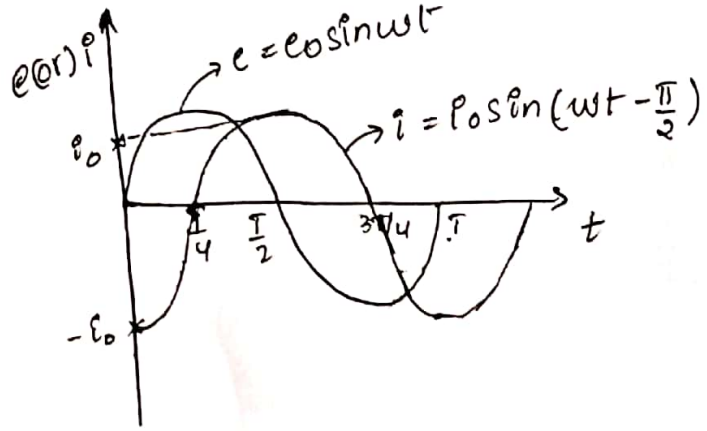
1. From eqns (1) & (2),

phase diff b|w e & i, $\phi = \frac{\pi}{2}$ rad

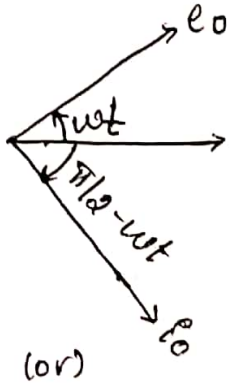
Clearly current "lags" emf by $\frac{\pi}{2}$ rad.

∴ power factor $= \cos\phi = \cos\frac{\pi}{2} = 0$.

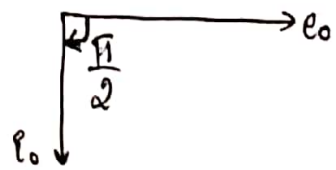
2. wave form:



phasor diagram:



(or)



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Q. we have,

$$I_0 = \frac{e_0}{\omega L} = \frac{e_0}{X_L}$$

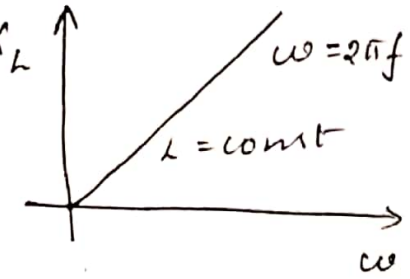
where $X_L = \omega L$ = inductive reactance of circuit.

X_L - resistance offered by inductor in AC circuit is called inductive reactance.

unit : ohm (Ω).

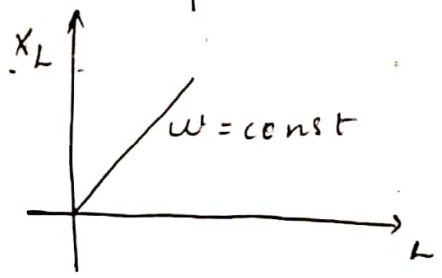
** $X_L \propto \omega$ (where L = const)

$X_L \propto L$ (where ω = const)



For D.C, $f = 0$

$\Rightarrow \boxed{X_L = 0}$

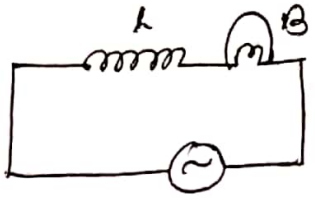


At low frequency, resistance (reactance, X_L) is less.

∴ Inductance opposes high frequency currents and can pass low frequency currents easily.

* It acts as "low pass filter".

Exe



as $f \uparrow \Rightarrow X_L$ also increases

$$I_0 = \frac{e_0}{X_L}$$

as $X_L \uparrow \Rightarrow I_0 \downarrow$

∴ Brightness of bulb decreases.

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* instantaneous power,

$$P_{ins} = ei = e_0 \sin\omega t \; i_0 \sin(\omega t - \tfrac{\pi}{2})$$

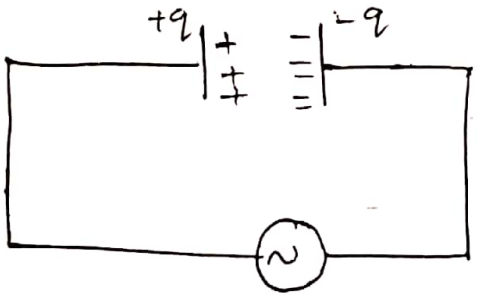
$$= -\frac{1}{2} e_0 i_0 \sin 2\omega t$$

* Avg power,

$$\langle P \rangle = \frac{1}{2} e_0 i_0 \cos\phi = 0.$$

$$X_L = \frac{e_0}{i_0} = \frac{\sqrt{2}\, e_{rms}}{\sqrt{2}\, i_{rms}} = \frac{e_{rms}}{i_{rms}} = \omega L$$

AC circuit containing pure capacitor only :



$$e = e_0 \sin\omega t \quad -(1)$$

At any time t,

$$e - \frac{q}{C} = 0$$

$$q = eC$$

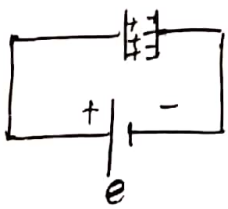
$$q = C e_0 \sin\omega t$$

$$i = \frac{dq}{dt} = C e_0 \omega \cos\omega t$$

$$\therefore \quad i = i_0 \sin(\omega t + \tfrac{\pi}{2})$$

where $i_0 = e_0 \omega C$

D.C circuit :



$$e = \frac{q}{C}$$

$$q = CV$$

$$i = \frac{dq}{dt} = 0.$$

Capacitor blocks DC.

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$$i_0 = \frac{e_0}{\frac{1}{\omega c}} = \text{peak current of the circuit.}$$

points:

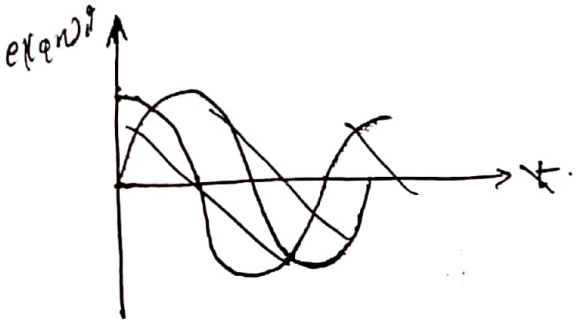
1. phase difference b/w emf and current,

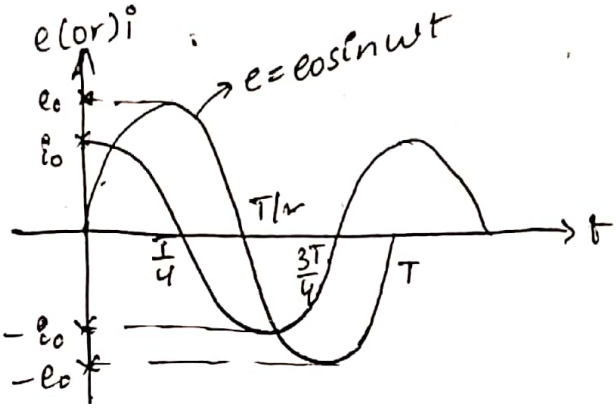
$$\phi = \frac{\pi}{2} \text{ rad}$$

clearly current "leads" emf by $\frac{\pi}{2}$ rad.

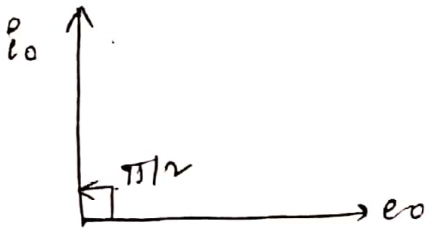
$\Rightarrow$ power factor $= \cos\phi = \cos\frac{\pi}{2} = 0$.

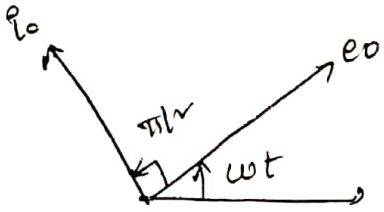
2. wave form:





phasor diagram:





3. we have,

$$i_0 = \frac{e_0}{\frac{1}{\omega c}} = \frac{e_0}{X_c}$$

where, $X_c = \frac{1}{\omega c}$ = capacitive reactance. (ohm)

clearly,

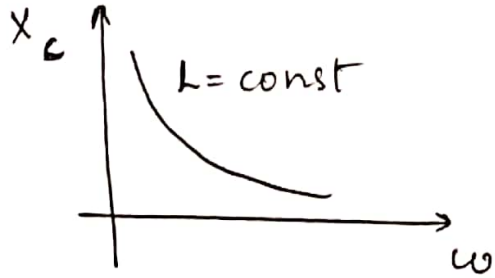
(i) For DC, $f = 0 \Rightarrow X_c \to$ infinity.

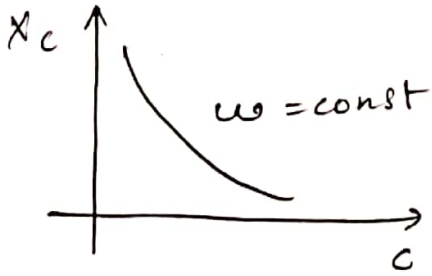
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Here capacitor block Dc.

(ii) $X_C \propto \frac{1}{\omega}$ (C = const)

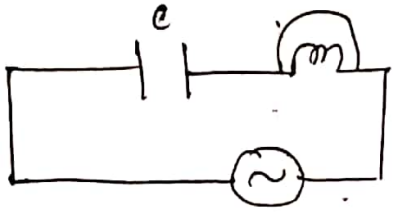
$X_C \propto \frac{1}{C}$ (ω = const)





* capacitor acts as "high pass filter".

Ex6



As $C = \frac{\varepsilon_0 A}{d}$

if $d \downarrow \Rightarrow C \uparrow$.

$$X_C = \frac{1}{\omega C} \Rightarrow X_C \downarrow$$

$$i_0 = \frac{e_0}{X_C} \Rightarrow i_0 \uparrow$$

∴ Brightness of bulb increases.

* Instantaneous power,

$$P_{ins} = ei = e_0 \sin\omega t \; i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$$

$$P_{ins} = \frac{1}{2} e_0 i_0 \sin(2\omega t)$$

* Avg Power,

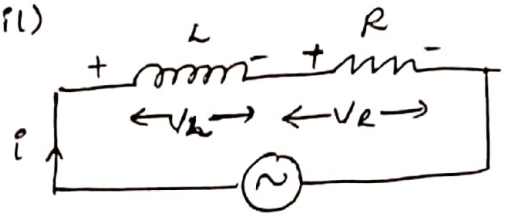
$$\langle P \rangle = \frac{1}{2} e_0 i_0 \cos\phi = 0$$

***

$$\boxed{X_C = \frac{1}{\omega C} = \frac{e_0}{i_0} = \frac{e_{rms}}{i_{rms}}}$$

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AC circuit containing Inductance (L) & resistance (R) in series (choke coil)



using KVL,

$$e - L\frac{di}{dt} - iR = 0$$

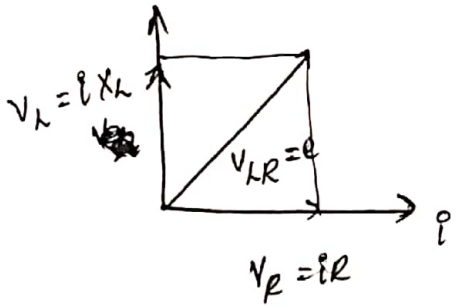
$$e = e_0 \sin\omega t \quad -(1)$$

Let the eqn of current is,

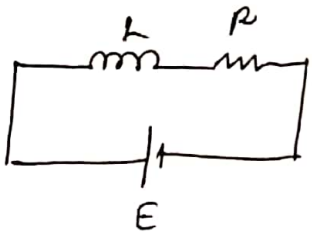
$$i = i_0 \sin(\omega t - \phi)$$

here i_0 & ϕ — unknowns.

Phasor diagram



DC circuit



From figure,

$$e = \sqrt{V_R^2 + V_L^2}$$

$$e = \sqrt{(iR)^2 + (iX_L)^2}$$

$$e = i\sqrt{R^2 + X_L^2}$$

$$\therefore \frac{e}{i} = \sqrt{R^2 + X_L^2} = Z_{LR}$$

Z_{LR} - impedance of LR circuit.

↓

units : ohm (Ω)

* $\sqrt{X_L^2 + R^2} = Z_{LR}$ — resistance offered by L-R circuit to

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AC circuit is called Z_{LR}.

* Reciprocal of impedance is called "Admittance" (Y)

* units of Y — Ω^{-1} or mho or seimen.

* $Z_{LR} = \frac{e_0}{i_0} = \frac{e_{rms}}{i_{rms}}$

$$i_0 = \frac{e_0}{Z_{LC}} = \frac{e_0}{\sqrt{R^2 + X_L^2}}$$

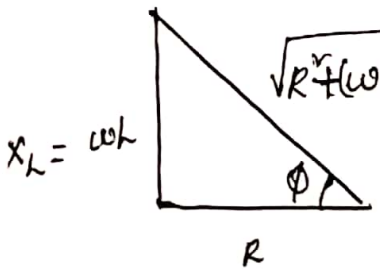
$$\boxed{i_0 = \frac{e_0}{\sqrt{R^2 + (\omega L)^2}}} \star\star$$

and $\tan\phi = \frac{V_L}{V_R} \frac{i X_L}{iR}$

$$\tan\phi = \frac{X_L}{R}$$

$$\tan\phi = \frac{\omega L}{R}$$

$$\boxed{\phi = \tan^{-1}\left(\frac{\omega L}{R}\right)} \star$$



$$\sin\phi = \frac{X_L}{Z_{LR}}$$

$$\boxed{\cos\phi = \frac{R}{Z_{LR}} = \text{power factor}}$$

* $Z_{LR} = \frac{e_0}{i_0} = \frac{e_{rms}}{i_{rms}}$

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* $P_{instantaneous}$,

$$P_{inst} = ei = e_0 i_0 \sin\omega t \sin(\omega t - \phi)$$

* Avg power,

$$\langle P \rangle = \frac{1}{2} e_0 i_0 \cos\phi$$

$$= \frac{1}{2} e_0 i_0 \frac{R}{Z_{LR}}$$

$$\boxed{\langle P \rangle = \frac{e_0^2 R}{2 Z_{LR}^2}}$$

* energy consumed in the circuit in a time 't',

$$E = \langle P \rangle t = \frac{e_0^2 R t}{2 Z_{LR}^2}$$

* Seperately,

$$V_R = V_{R0} \sin(\omega t - \phi) \quad \text{where, } V_{R0} = i_0 R = \frac{e_0 R}{Z_{LR}}$$

$$\phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

* Voltage drop across inductor,

$$V_L = V_{L0} \sin\left(\omega t - \phi + \frac{\pi}{2}\right)$$

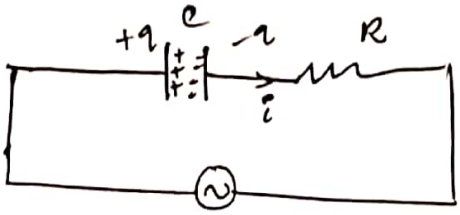
where, $V_{L0} = i_0 X_L$

$$= i_0 (\omega L)$$

$$V_{L0} = \frac{e_0 (\omega L)}{Z_{LR}}$$

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AC circuit containing capacitance (C) and resistance (R) in series:



At any time 't',

$$e - \frac{q}{C} - iR = 0 \qquad e = e_0 \sin \omega t \quad - (1)$$

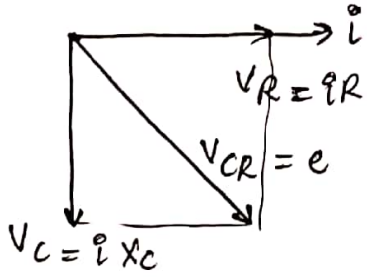
Let eqn of current,

$$i = i_0 \sin(\omega t + \phi)$$

here i_0, ϕ – unknowns

To find i_0, ϕ – use phasor diagram.

Phasor diagram:



$$e = \sqrt{V_R^2 + V_C^2}$$

$$= \sqrt{(iR)^2 + (i X_C)^2}$$

$$e = i\sqrt{R^2 + X_C^2}$$

$$\frac{e}{i} = \sqrt{R^2 + X_C^2}$$

$$\boxed{Z_{CR} = \sqrt{R^2 + X_C^2}}$$

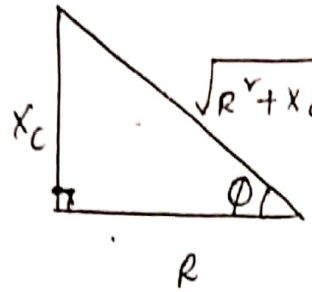
Z_{CR} – impedance of CR circuit. (Ω)

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$$Z_{CR} = \frac{e_0}{i_0} = \frac{e_{rms}}{i_{rms}} = \sqrt{R^2 + X_C^2}$$

and $\tan\phi = \frac{V_C}{V_R} = \frac{i X_C}{i R}$

$$\tan\phi = \frac{1}{\omega C R}$$



$$\sin\phi = \frac{X_C}{Z_{CR}}$$

$$\cos\phi = \frac{R}{Z_{CR}} = \text{power factor}$$

* Instantaneous power,

$$P_{inst} = e i.$$

* Avg power,

$$\langle P \rangle = \frac{1}{2} e_0 i_0 \cos\phi$$

$$\langle P \rangle = \frac{e_0^2 R}{2 Z_{CR}^2}$$

* Energy consumed,

$$E = P_{avg}\, t$$

$$E = \frac{e_0^2 R}{2 Z_{CR}^2} t$$

* Seperately,

Voltage drop across resistance 'R' $\Rightarrow V_R = V_{R0} \sin(\omega t + \phi)$ where, $V_{R0} = i_0 R$

$\phi = \tan^{-1}\left(\frac{1}{\omega C R}\right)$

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Voltage drop across capacitance (C),

$$V_C = V_{C0} \sin(\omega t + \phi - \tfrac{\pi}{2})$$

where, $V_{C0} = I_0 X_C = \dfrac{I_0}{\omega C}$.

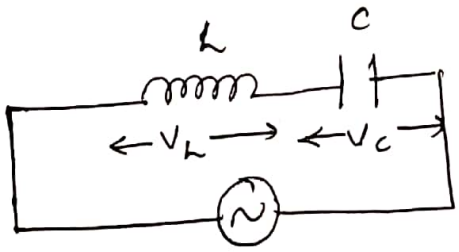
charge on plates of capacitance,

$$q = C V_{C0} \sin(\omega t + \phi - \tfrac{\pi}{2})$$

$$= \cancel{C} \dfrac{I_0}{\omega \cancel{C}} \sin(\omega t + \phi - \tfrac{\pi}{2})$$

$$\boxed{q = \frac{I_0}{\omega} \sin(\omega t + \phi - \tfrac{\pi}{2})}$$

AC circuit containing capacitance (C) & inductance (L) connected in series :



at any time 't',

$$e - L \cdot \frac{di}{dt} - \frac{q}{C} = 0 \qquad e = e_0 \sin\omega t \quad - (1)$$

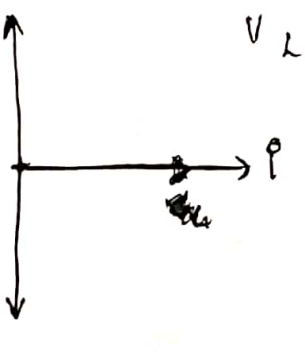
Let the eqn of current is

$$q = I_0 \sin(\omega t \pm \phi)$$

here, I_0 & ϕ are unknowns.

phasor diagrams

case (1): V_L

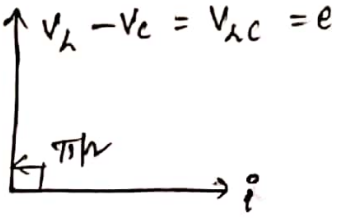


$V_L > V_C \Rightarrow X_L > X_C$

here circuit is inductive (at high frequencies)

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$\Rightarrow$



$$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$$

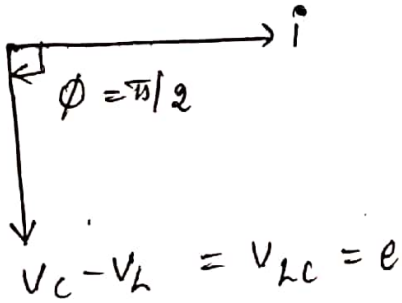
$$e = V_L - V_C$$

$$e = i(X_L - X_C)$$

$$\frac{e}{i} = (X_L - X_C) = X_{LC} = \text{net reactance of circuit.}$$

$\therefore$ $$\boxed{i_0 = \frac{e_0}{X_L - X_C}}$$

case (2): $V_L < V_C \Rightarrow X_L < X_C$



here circuit is capacitive (at low frequencies)

$$i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$$

where $$i_0 = \frac{-e_0}{X_C - X_L}$$

case (3): At a particular frequency, $\boxed{\omega = \omega_0}$

here $X_L = X_C$

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{LC}}} \rightarrow \text{Resonant frequency.}$$

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net reactance, $\boxed{X = 0}$

current, $I_0 = \frac{e_0}{X}$

I_0 = infinity.

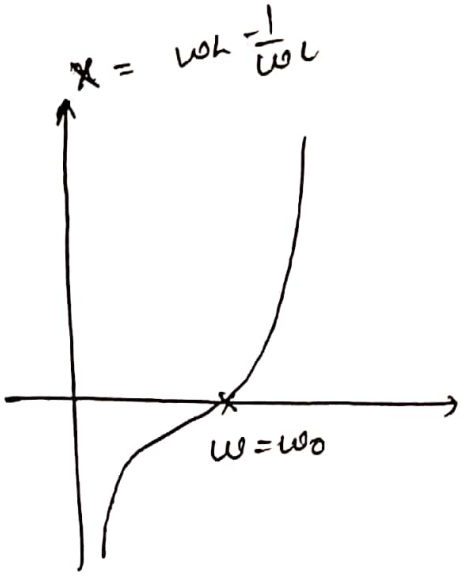
Graphs:

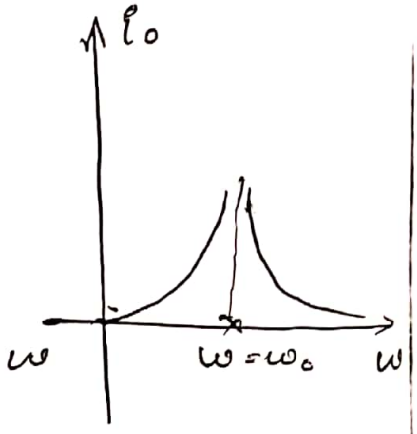
1) $I_0 = \frac{e_0}{|X|} = \frac{e_0}{|\omega L - \frac{1}{\omega C}|}$

$\omega \to 0 \Rightarrow I_0 \to 0$

$\omega \to \infty \Rightarrow I_0 \to 0$

$\omega \to \omega_0 \Rightarrow I_0 \to \infty$





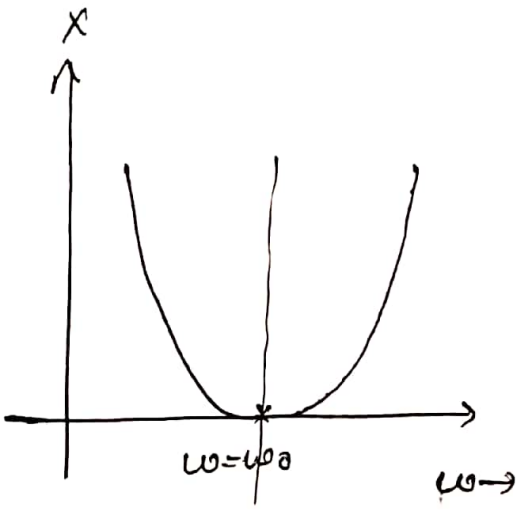
2) Net reactance,

$|X| = |\omega L - \frac{1}{\omega C}|$

$\omega \to 0 \Rightarrow X \to \infty$

$\omega \to \infty \Rightarrow X \to \infty$

$\omega \to \omega_0 \Rightarrow X = 0$



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Ex8 The voltage applied to a purely inductive coil of self inductance 15.9 mH is given by eqn $V = 100\sin 314t + 75\sin 942t + 50\sin 1570t$. Find the eqn of resulting current wave.

Sol8

Comparing with $V_{o1}\sin\omega_1 t + V_{o2}\sin\omega_2 t + V_{o3}\sin\omega_3 t$

$$\Rightarrow \quad V_{o1} = 100 \;, V_{o2} = 75, \; V_{o3} = 50$$

$$\omega_1 = 314 \;, \; \omega_2 = 942, \; \omega_3 = 1570$$

$$I_{o1} = \frac{V_{o1}}{\omega_1 L} \;, \; I_{o2} = \frac{V_{o2}}{\omega_2 L} \;, \; I_{o3} = \frac{V_{o3}}{\omega_3 L}$$

Finally, $i = I_{o1}\sin\left(\omega_1 t - \frac{\pi}{2}\right) + I_{o2}\sin\left(\omega_2 t - \frac{\pi}{2}\right) + I_{o3}\sin\left(\omega_3 t - \frac{\pi}{2}\right)$

Solve =)

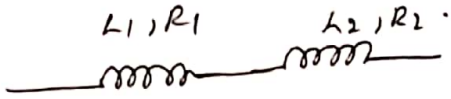
Ex8 Two coils are connected in series with 2A DC through the circuit. The potential diff across the coils are 20V & 30V respectively with 2A AC at 40Hz. The potential diff across coils are 140V & 100V respectively. The 2 coils are connected in series with a 230V, 50Hz supply. Find (i) current in the coils.

(ii) power factor of the circuit.

(iii) power dissipated in the circuit.

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Sol⁶



with 2A DC,

$$R_1 = \frac{V1}{I} = \frac{20}{2} = 10\,\Omega$$

$$R_2 = \frac{V2}{I} = \frac{30}{2} = 15\,\Omega.$$

with 2A AC,

$$Z_1 = \frac{V_{rms}}{I_{rms}} = \frac{140}{2} = 70\,\Omega$$

$$Z_2 = \frac{V_{rms}}{I_{rms}} = \frac{100}{2} = 50\,\Omega$$

now, $Z_1 = \sqrt{R_1^2 + \omega_1^2 L_1^2}$, $Z_2 = \sqrt{R_2^2 + \omega_1 L_2^2}$

$\omega_1 = 2\pi f = 2\pi(40) = 80\pi$ $\Rightarrow L_1 , L_2 = ?$.

now, at $\omega_2 = 8\pi \times 50 = 100\pi$.

$\Rightarrow Z_1 = \sqrt{R_1^2 + \omega_2 L_1^2} =$

$Z_2 = \sqrt{R_2^2 + \omega_2 L_2^2} =$

Finally, $I_{rms} = \frac{V_{rms}}{Z_1 + Z_2}$

power factor $= \frac{R}{Z} = \frac{R_1 + R_2}{Z_1 + Z_2}$

power consumed = I_{rms} V_{rms} $\cos\phi$

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Ex 6 given $i = 2\sin 100\pi t + 2\sin\left(100\pi t + \frac{\pi}{6}\right)$ then find i_{eff}.

Sol6

$$i_0 = \sqrt{i_1^2 + i_2^2 + 2i_1 i_2 \cos\phi}$$

$$= \sqrt{4 + 4 + 2 \times 4 \times \frac{\sqrt{3}}{2}}$$

$$= \sqrt{8 + 4\sqrt{3}}$$

$$i_{rms} = \frac{i_0}{\sqrt{2}} = \sqrt{\frac{8 + 4\sqrt{3}}{2}} = \sqrt{4 + 2\sqrt{3}}$$

$$\boxed{i_{rms} = 2 + \sqrt{3}}$$

* choke coil $\Rightarrow$ $L \uparrow \Rightarrow R \downarrow$

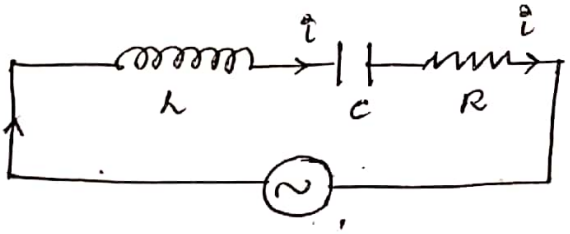
$P_{loss} \approx 0$, Power factor $= \frac{R}{Z} = 0$

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ALTERNATING CURRENT (continuation)

2|07|25 | wednesday

L-C-R series circuit:



$e = e_0 \sin \omega t$

At any time t,

$$e - L\cdot\frac{di}{dt} - \frac{q}{C} - iR = 0$$

Let 'i' be the current in the circuit,

$i = i_0 \sin(\omega t \pm \phi)$

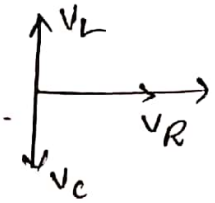
⟶ unknowns are i_0, ϕ

phasor diagram:

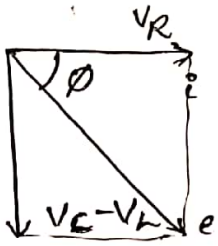
1) At low frequencies, $X_L < X_C \Rightarrow V_L < V_C$

Here circuit is capacitive.

originally:



on this case:



$$e = \sqrt{V_R^2 + (V_C - V_L)^2}$$

$$iZ = i\sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Z — impedance of circuit

and peak current, $i = \frac{e_0}{Z} = \frac{e_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

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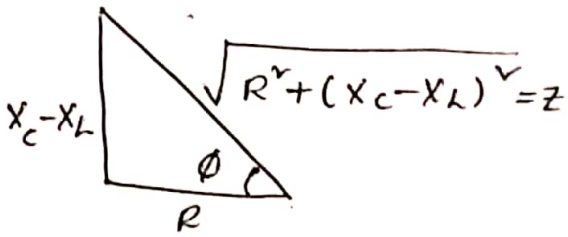
$$Z = \frac{e_0}{i_0} = \frac{e_{rms}}{i_{rms}}$$

* current eqn,

$$i = i_0 \sin(\omega t + \phi)$$

where $\tan\phi = \frac{V_C - V_L}{V_R}$

$$\tan\phi = \frac{X_C - X_L}{R}$$



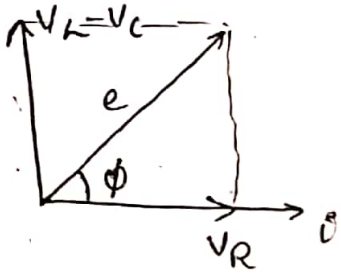
$$\cos\phi = \frac{R}{Z}$$ - power factor.

* $P_{avg} = \frac{1}{2} i_0 e_0 \cos\phi$

$$P_{avg} = \frac{e_0^2 R}{2 Z^2}$$

2) At high frequencies, $X_L > X_C \Rightarrow V_L > V_C$

Here circuit is inductive.



$$e = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

current eqn, $i = i_0 \sin(\omega t - \phi)$

$$i_0 = \frac{e_0}{\sqrt{R^2 + (X_L - X_C)^2}}$$

Power factor $\Rightarrow \cos\phi = \frac{R}{Z}$

phase $\Rightarrow \phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

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3) At one particular frequency,

$$\omega = \omega_0$$

$$X_L = X_C$$

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$ resonant frequency (rad/sec)

(or)

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{LC}}}\ (Hz)$$

clearly resonant frequency depends on L-& C values but not on R value.

* Here, net reactance,

$$X = X_L - X_C = 0$$

impendence of circuit,

$$\boxed{Z = R}\ (min)$$

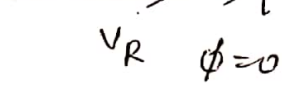
* peak current,

$$i_0 = \frac{e_0}{Z} = \frac{e_0}{R}\ (max)$$

* Clearly i_0 (max peak current) depends on R values but not on L & C value

* phase diff b/w emf & current,

$$\boxed{\phi = 0}$$



* power factor, $\cos\phi = 1$ (max)

* Avg power consumed,

$$\langle P \rangle = \frac{1}{2} e_0 i_0 \cos\phi$$

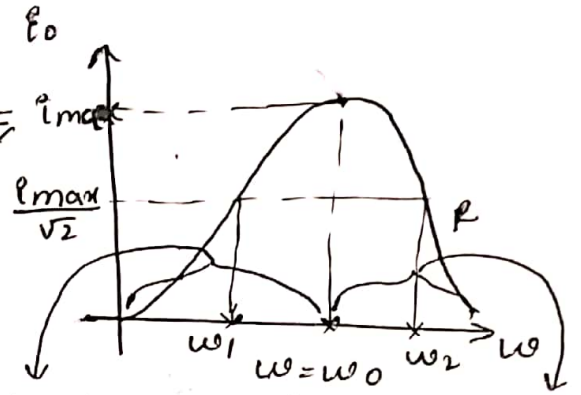
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$$P_{avg} = \frac{e_0^2}{2R} = \frac{1}{2} i_0^2 R$$

Graphs

① current amplitude,

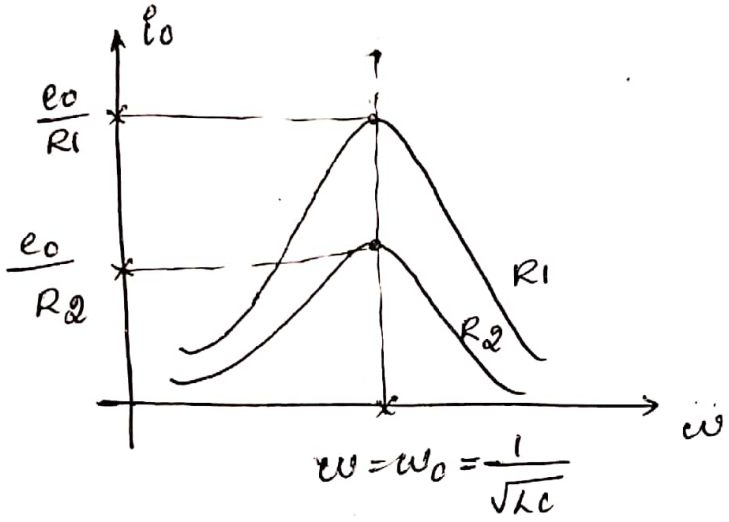
$$i_0 = \frac{e_0}{Z} = \frac{e_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$



$\omega \to 0 \Rightarrow i_0 \to 0$

$\omega \to \infty \Rightarrow i_0 \to 0$

$\omega \to \omega_0 \Rightarrow i_0 \to \frac{e_0}{R}$



$$\frac{e_0}{R_1} > \frac{e_0}{R_2}$$

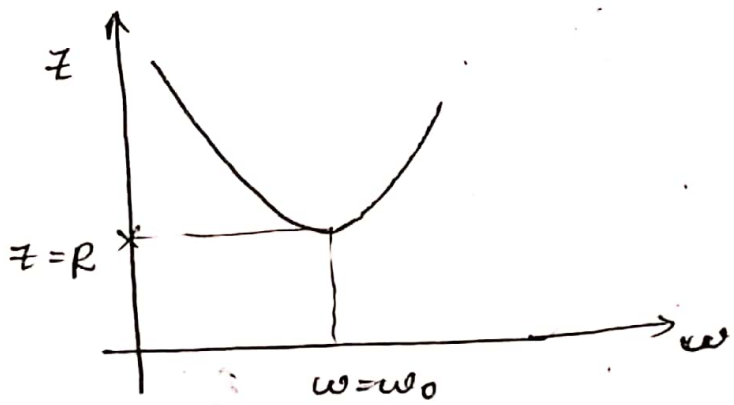
$$\boxed{R_2 > R_1}$$

② Impedence, $Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$

$\omega \to 0 \Rightarrow Z \to \infty$

$\omega \to \infty \Rightarrow Z \to \infty$

$\omega \to \omega_0 \Rightarrow Z_0 = R$ (min)



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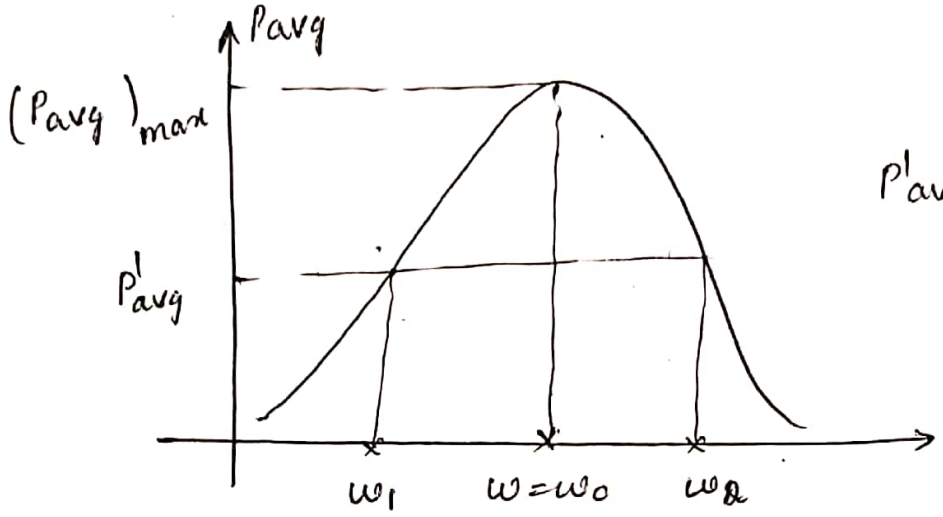
③ Average power consumed,

$$P_{avg} = \frac{e_0^2 R}{2Z^2} = \frac{e_0^2 R}{2\left(R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right)}$$

at $\omega \to 0 \Rightarrow P_{avg} \to 0$

at $\omega \to \infty \Rightarrow P_{avg} \to 0$

at $\omega \to \omega_0 \Rightarrow P_{avg} = \frac{e_0^2}{2R} = \frac{1}{2} i_0^2 R \ (max)$



$P'_{avg} = \frac{1}{2}(P_{avg})_{max}$

$\omega_1, \omega_2 \longrightarrow$ half power frequencies.

$\Delta\omega = \omega_2 - \omega_1 =$ band width.

$$\frac{1}{2} i^2 R = \frac{1}{2} \times \frac{1}{2} i_0^2 R \Rightarrow \boxed{i = i_0/\sqrt{2}}$$

* The frequencies at which power consumed in the circuit is equal to half of the max power consumed at resonance are called "Half power frequency"

* Again,

$$\frac{e_0^2 R}{2\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]} = \frac{1}{2} \times \frac{e_0^2}{2R}$$

$$2R^2 = R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$R^2 = \left(\omega L - \frac{1}{\omega C}\right)^2$$

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$$\omega L - \frac{1}{\omega C} = \pm R$$

$$\Rightarrow \omega_1 L - \frac{1}{\omega_1 C} = R \quad \text{— (a)}$$

$$\Rightarrow \omega_2 L - \frac{1}{\omega_2 C} = -R \quad \text{— (b)}$$

Adding eqns (a) & (b)

$$(\omega_1 + \omega_2) L - \frac{1}{C}\left(\frac{1}{\omega_1} + \frac{1}{\omega_2}\right) = 0$$

$$L = \frac{1}{\omega_1 \omega_2 C}$$

$$\boxed{\omega_1 \omega_2 = \frac{1}{LC}}$$

and eqn (b) − (a)

$$(\omega_2 - \omega_1) L + \left(\frac{1}{\omega_1} - \frac{1}{\omega_2}\right)\frac{1}{C} = 2R$$

$$(\omega_2 - \omega_1)\left(L + \frac{1}{\omega_1 \omega_2 C}\right) = 2R$$

$$(\omega_2 - \omega_1)\left(\cancel{2}L\right) = \cancel{2}R$$

$$\boxed{\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L}} = \text{bandwidth (rad/sec)}$$

& $f_2 - f_1 = \frac{R}{2\pi L}$ (Hz)

Quality Factor :

1) $Q = \dfrac{\text{Resonant frequency}}{\text{band width}} = \dfrac{\omega_0}{\Delta\omega} = \dfrac{1}{\sqrt{LC}}\left(\dfrac{L}{R}\right)$

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$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

It has no units and dimensions.

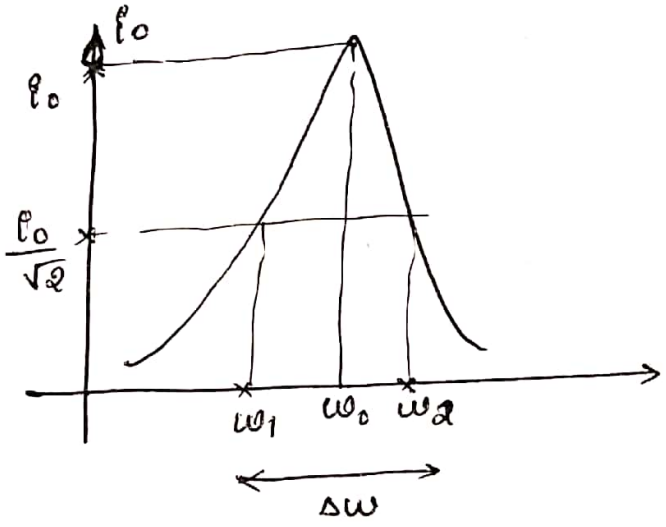
To have more Q values, R ↓, C ↓, L ↑.

2) At resonance,

a) $Q = \frac{V_L}{V_R} = \frac{I \omega_0 L}{I R} = \frac{\omega_0 L}{R}$

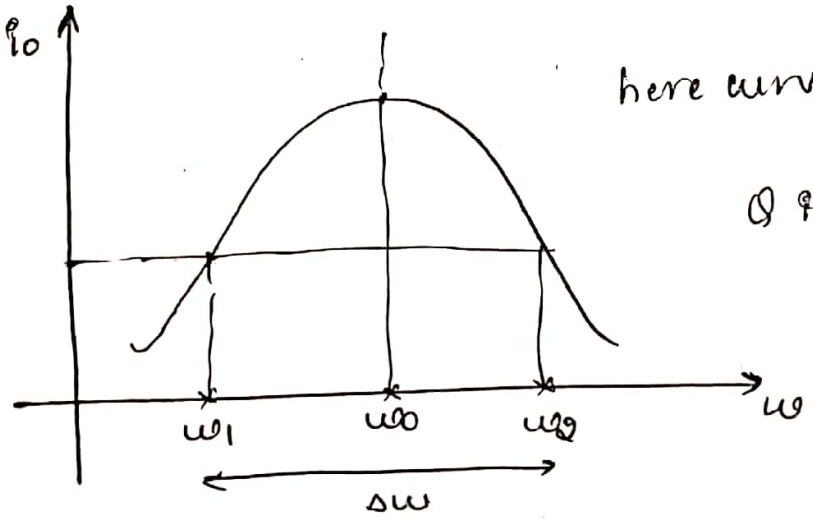
b) $Q = \frac{V_C}{V_R} = \frac{I}{\omega_0 C R (I)} = \frac{1}{\omega_0 CR}$

* Q is high then circuit has more sharpness of resonance.



Q is high

curve is more sharper.

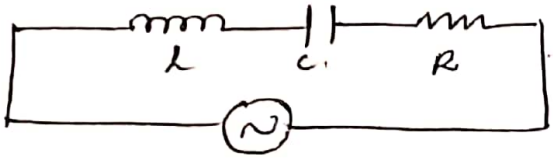


here curve is flater.

Q is low.

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* L-C-R circuit (resonant) is called as "acceptor circuit".



$$e = e_0 \sin \omega t$$

$$i = i_0 \sin(\omega t + \phi)$$

$$V_R = V_{R0} \sin(\omega t + \phi)$$

$$\phi = \tan^{-1}\left(\frac{X_C - X_L}{R}\right)$$

$$V_{R0} = i_0 R$$

$$V_C = V_{C0} \sin(\omega t + \phi - 90^\circ)$$

$$V_{C0} = i_0 X_L$$

$$V_L = V_{L0} \sin(\omega t + \phi + 90^\circ)$$

$$V_{L0} = i_0 X_L$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$P_{avg} = \frac{1}{2} e_0 i_0 \cos\phi$$

Ex 1) A 750Hz, 20V source is connected to a resistance of 100Ω, and inductance of 0.1803H and a capacitance of 10μF in series. Calculate the time in which the resistance which as thermal capacity 2J/°C will get heated by 10°C.

348.15sec

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2) A choke coil is needed to operate an arc lamp at 160V, 50Hz AC supply. The arc lamp has an R eff of 5Ω, when running at 10A. calculate the inductance of the choke coil. If the same arc lamp is to be operated on 160V DC supply, what additional resistance is required. Compare power loss in both cases.

Sol6 g) $V_{rms} = I_{rms} \cdot Z$

$$Z = 16 \Rightarrow R^2 + X_L^2 = Z^2$$

$$X_L = 15.2$$

$$\omega L = 15.2$$

$$\boxed{L = 0.05H}$$

$$I = \frac{V}{R_{eff}}$$

$$R + R' = 16$$

$$\boxed{R' = 11\Omega}$$

$$P_{avg_{(ac)}} = \frac{1}{2} e_0 i_0 \cos\phi$$

$$= \frac{1}{2} \times 160 \times \frac{10}{\sqrt{2}} \times 10 \times \frac{5}{\sqrt{2}} \times \frac{5}{16}$$

$$\boxed{P_{avg_{(ac)}} = 500\,J}$$

$$P_{loss_{(dc)}} = i^2 R = 100 \times 16$$

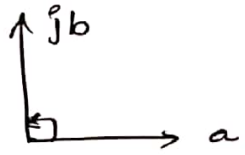
$$\boxed{P_{loss\,(dc} = 1600\,J}$$

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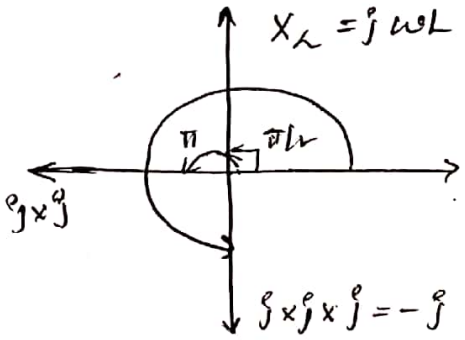
Complex Numbers

$z = a + jb$, $j = \sqrt{-1}$

$|z| = \sqrt{a^2 + b^2}$



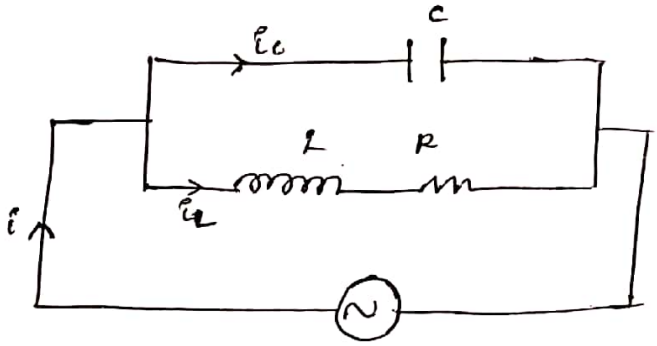
*



$$X_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

Ex6

Paallel rejector circuit :-



Find impedance of the circuit,

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$\frac{1}{Z} = \frac{1}{\frac{-j}{\omega C}} + \frac{1}{R + j\omega L}$$

$$\frac{1}{Z} = j\omega C + \frac{1}{R + \omega L j} \times \frac{R - j\omega L}{R - j\omega L}$$

$$\frac{1}{Z} = j\omega C + \frac{R - j\omega L}{R^2 + \omega^2 L^2}$$

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Admittance,

$$Y = \frac{1}{Z} = \frac{R}{R^2 + \omega^2 L^2} + j\left(\omega C - \frac{\omega L}{R^2 + \omega^2 L^2}\right)$$

$$|Y| = \left|\frac{1}{Z}\right| = \sqrt{\left(\frac{R}{R^2 + \omega^2 L^2}\right)^2 + \left(\omega C - \frac{\omega L}{R^2 + \omega^2 L^2}\right)^2}$$

At one frequency, (Resonant frequency)

$$\omega_0 C = \frac{\omega_0 L}{R^2 + \omega_0^2 L^2}$$

$$\frac{1}{Z} = Y = \frac{R}{R^2 + \omega_0^2 L^2} = \min$$

$$Z = \frac{R^2 + \omega_0^2 L^2}{R} \min$$

$$R^2 C + \omega_0^2 L^2 C = L$$

In resonance,

* current through capacitor,

$$I_C = \frac{V_{rms}}{X_C} = (V_{rms})(\omega_{res})(C)$$

* current through inductor/ resistor,

$$I_L = \frac{V_{rms}}{Z_L} = \frac{V_{rms}}{\sqrt{R^2 + \omega_{res}^2 L^2}}$$

$$= \frac{V_{rms}}{\sqrt{R^2 + \left(\frac{1}{LC} - \frac{R^2}{L^2}\right)L^2}}$$

$$I_L = V_{rms}\sqrt{\frac{C}{L}}$$

$$\omega_0^2 L^2 = L - R^2 C$$

$$\omega_0^2 = \frac{L - R^2 C}{L^2 C}$$

$$\omega_0^2 = \frac{1}{LC} - \frac{R^2}{L^2}$$

$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

↳ resonant frequency, depends on L, C, R.

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$$Z_{max} = R + \frac{L^2}{R}\left(\frac{1}{LC} - \frac{R^2}{L^2}\right)$$

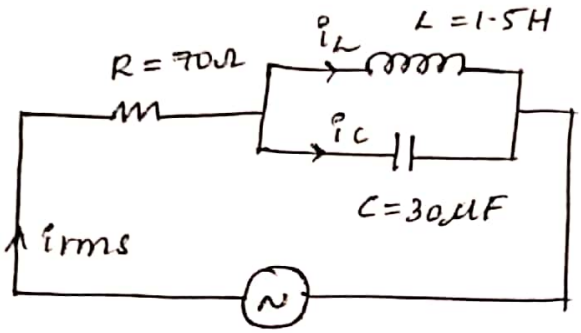
$$= R + \frac{L}{RC} - R$$

$$Z_{max} = \frac{L}{RC} = \text{dynamic resistance.}$$

Here min current in the circuit,

$$i_{min} = \frac{V_{rms}}{Z_{max}}$$ (at resonant frequency)

Ex 6



230V, 300 rad/s

$i_{rms} = 1.41$

Find

① impedence of circuit. = 163.3

② Find Resonant frequency of circuit.

③ rms value of total current.

④ current amplitude in the L & C arms of circuit.

$i_L = 0.496\ A$

$i_C = 1.87\ A$

⑤ power factor & power consumed in the circuit. $P_f = 0.43$

Sol 6

Impedance of L-C circuit,

$$\frac{1}{Z_1} = \frac{1}{\frac{1}{j\omega C}} + \frac{1}{j\omega L}$$

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$$\frac{1}{Z_1} = j\omega C + \frac{1}{j\omega L}$$

$$\frac{1}{Z_1} = \frac{-\omega^2 LC + 1}{j\omega L}$$

$$\boxed{Z_1 = \frac{j\omega L}{1-\omega^2 LC}} \Rightarrow \boxed{|Z_1| = \frac{\omega L}{1-\omega^2 LC}}$$

Total impedance of circuit,

$$Z = R + Z_1$$

$$Z = R + j\frac{\omega L}{1-\omega^2 LC}$$

Total impedence,

$$|Z| = \sqrt{R^2 + \left(\frac{\omega L}{1-\omega^2 LC}\right)^2}$$

At resonance,

$$1 = \omega_0^2 LC$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$ resonant frequency

here, $|Z| = \infty$ & $I_{rms} = \frac{V_{rms}}{Z} = 0$.

In general,

rms value of total current,

$$I_{rms} = \frac{V_{rms}}{Z} =$$

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$$I_{rms} = I_L + I_C$$

$$\text{i.e.} \quad I_L(j\omega L) = I_C\left(\frac{1}{j\omega C}\right)$$

$$I_L = \frac{-jI_C}{\omega^2 LC}$$

$$I_{rms} = I_C\left(1 - \frac{1}{\omega^2 LC}\right)$$

$$I_{rms} = I_C\left(\frac{\omega^2 LC - 1}{\omega^2 LC}\right)$$

$$\boxed{I_C = I_{rms}\left(\frac{\omega^2 LC}{\omega^2 LC - 1}\right)}$$

$$I_L = \frac{-1}{\omega^2 LC}\left(I_{rms}\left(\frac{\omega^2 LC}{\omega^2 LC - 1}\right)\right)$$

$$I_L = \frac{-I_{rms}}{\omega^2 LC - 1}$$

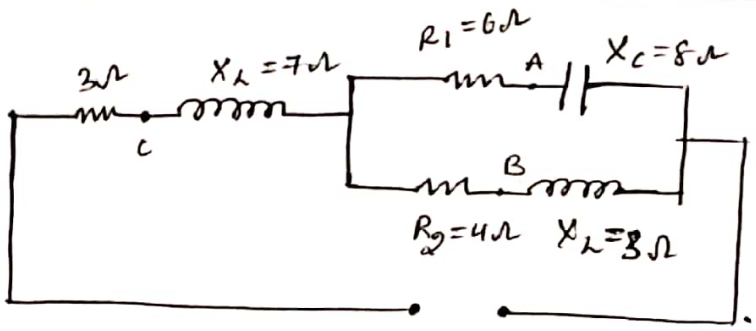
$$\boxed{I_L = \frac{I_{rms}}{1 - \omega^2 LC}}$$

current amplitudes, $I_{Co} = \sqrt{1}\, I_C$

$$I_{Lo} = \sqrt{2}\, I_L$$

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Ex 8



100V, 60Hz

Find impedance of circuit & rms value of total current.

Sol 8

$$Z = Z_0 + \frac{Z_A Z_B}{Z_A + Z_B}$$

$$= (3+7j) + \frac{(6-8j)(4+3j)}{(6-8j)+(4+3j)}$$

$$= 3+7j + \frac{24 - 24(-1) - 32j + 18j}{10 - 5j}$$

$$= 3+7j + \frac{48 - 14j}{10 - 5j}$$

$$= 3+7j + \frac{(48-14j)(10+5j)}{10^2+5^2}$$

$$= 3+7j + \frac{1}{125}(480 + 240j - 140j + 70)$$

$$= 3+7j + \frac{1}{125}(550 + 100j)$$

$$= 3+7j + \frac{1}{5}(22 + 4j)$$

$$= 3 + \frac{22}{5} + \left(7 + \frac{4}{5}\right)j = 3 + 4.4 + (7 + 0.8)j$$

$$Z = 7.4 + 7.8j$$

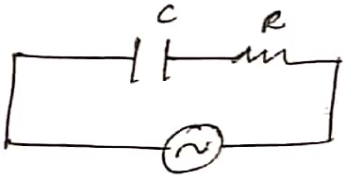
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$$|Z| = \sqrt{(7.4)^2 + (7.8)^2}$$

$$i_{rms} = \frac{V_{rms}}{|Z|}$$

$$=$$

Ex6 A voltage $V = 100\sin 314t$ is applied to a circuit consisting of a 25Ω resistor and 50μF capacitor in series.



Find 1) instantaneous current

2) power consumed in circuit 56.7

3) The p.d across the capacitor at the instant when the current is one half of its max value. 73.5 V

Sol6

1) $i = i_0 \sin(314t + \phi)$

$$i_0 = \frac{V_0}{Z} = \frac{100}{\sqrt{R^2 + X_c^2}}, \quad \phi = \cos^{-1}\left(\frac{R}{Z}\right)$$

2) power consumed,

$$P_{avg} = V_{rms} i_{rms} \cos\phi$$

3) Given $i = \frac{i_0}{2}$

$$\frac{1}{2} = \sin(314t + \phi)$$

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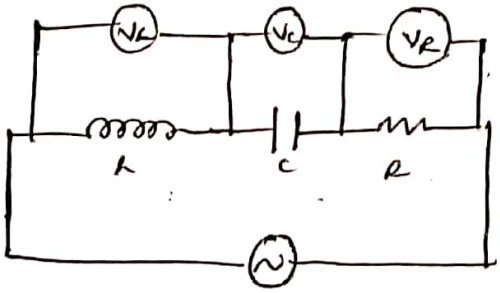
$314t + \phi = 30°, 150°$

$t =$

$V_c = V_{c_0} \sin(314t + \phi - 90°)$

$V_{c_0} = I_0 X_c$

Ex 6



$V = V_0 \sin\left(100t + \frac{\pi}{6}\right)$

$V_L = 40V$

$V_R = 40V$

$Z = 5\Omega$

$R = 4\Omega$

column - I		column - II	
A) peak current (A)	P	P)	$10\sqrt{2}$
B) V_0 (in volts)	Q	Q)	$50\sqrt{2}$
C) eff value of applied voltage (in volts)	R	R)	50
D) X_C (in Ω)	S	S)	1

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Sol6

$$V_R = (I_{rms})R$$

$$40 = I_{rms}(4)$$

$$\boxed{I_{rms} = 10}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} \Rightarrow I_0 = \sqrt{2}\, I_{rms} = 10\sqrt{2}$$

$$\boxed{I_0 = 14.14\ A}$$

we have,

$$V_{rms} = I_{rms}\, Z$$

$$V_{rms} = 10 \times 5$$

$$V_{rms} = 50V$$

$$V_0 = 50\sqrt{2}$$

$$\boxed{V_0 = 70.7V}$$

Finally,

$$V_{rms} = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$(50)^2 = (40)^2 + (40 - V_C)^2$$

$$90 \times 10 = (40 - V_C)^2$$

$$40 - V_C = 30$$

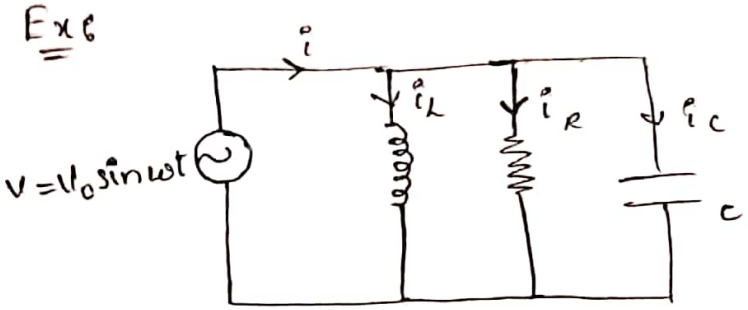
$$\boxed{V_C = 10V}$$

$$V_C = I_{rms} X_C$$

$$10 = 10\, X_C \Rightarrow \boxed{X_C = 1\Omega}$$

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Exc



To find impedence,

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{j\omega L} + \frac{1}{\frac{1}{j\omega C}}$$

$$\frac{1}{Z} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$

$$\frac{1}{Z} = \frac{1}{R} + \frac{1-\omega^2 LC}{j\omega L}$$

$$\frac{1}{Z} = \frac{1}{R} - j\left(\frac{1-\omega^2 LC}{\omega L}\right)$$

Admittance,

$$|Y| = \left|\frac{1}{Z}\right| = \sqrt{\frac{1}{R^2} + \left(\frac{1-\omega^2 LC}{\omega L}\right)^2}$$

At resonance,

$$\omega_0 C = \frac{1}{\omega_0 L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$Y = \frac{1}{R} = \frac{1}{Z} \rightarrow$ minimum.

$\Rightarrow$ impedance,

$Z = R$ (max)

Here current from supply,

$$i_0 = \frac{V_0}{Z} = \frac{V_0}{R} \text{ (max)}$$

currents through i, i_L, i_R, i_C are

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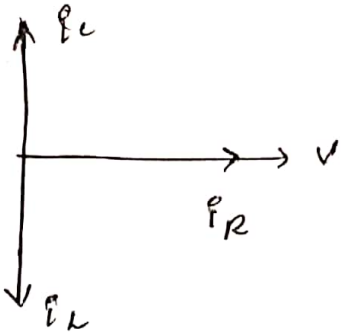
1) $i_L = \frac{V_{rms}}{X_L} = \frac{V_{rms}}{\omega_0 L}$

2) $i_R = \frac{V_{rms}}{R}$

3) $i_C = \frac{V_{rms}}{X_C} = \omega_0 C V_{rms}$

* current from supply,

$$i_{rms} = \sqrt{i_R^2 + (i_C - i_L)^2}$$



phase difference b/w current and emf is, $\tan\phi = \frac{i_C - i_L}{i_R}$

$$\boxed{\phi = \tan^{-1}\left(\frac{i_C - i_L}{i_R}\right)}$$

instantaneous currents,

$$i_L = i_{L_0} \sin(\omega t - \tfrac{\pi}{2}) \quad ; \quad i_{L_0} = \frac{V_0}{X_L}$$

$$i_R = i_{R_0} \sin(\omega t) \quad ; \quad i_{R_0} = \frac{V_0}{R}$$

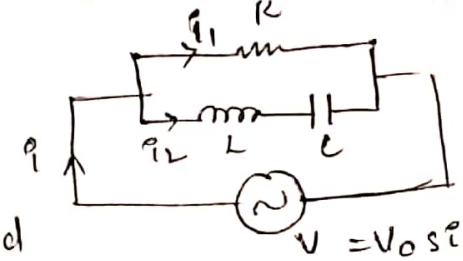
$$i_C = i_{C_0} \sin(\omega t + \pi/2) \quad ; \quad i_{C_0} = \frac{V_0}{X_C}$$

P_{avg} consumed,

$$\langle P \rangle = V_{rms}\, i_{rms} \cos\phi .$$

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Ex 6



Find $V = V_0 \sin \omega t$

1) Resonant frequency.

2) Impedance of the circuit.

Sol 6

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{R}$$

$$Z_1 = j\omega L - \frac{j}{\omega C}$$

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{j\omega L - \frac{j}{\omega C}}$$

$$\frac{1}{Z} = \frac{1}{R} - \frac{j\omega C}{\omega^2 LC - 1}$$

Admittance,

$$|Y| = \left|\frac{1}{Z}\right| = \sqrt{\frac{1}{R^2} + \left(\frac{\omega C}{\omega^2 LC - 1}\right)^2}$$

At $\omega_0 = \frac{1}{\sqrt{LC}}$,

(resonant frequency)

$$|Y| = \left|\frac{1}{Z}\right| = \infty \quad \Rightarrow \quad |Z| = 0$$

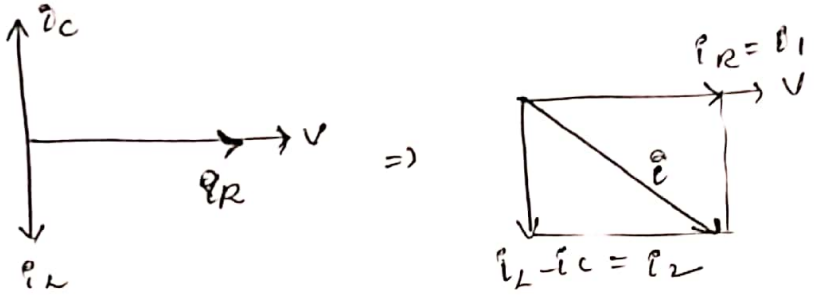
$$i_{rms} = \frac{V_{rms}}{Z} = \infty.$$

Here,

$$i_1 = \frac{V_{rms}}{R}$$

$$i_2 = \frac{V_{rms}}{X} = \frac{V_{rms}}{\left|\omega L - \frac{1}{\omega C}\right|}$$

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Here, $\boxed{i = \sqrt{i_1^2 + i_2^2}}$ *

$V_C = i_2 (X_C)$ $\quad$ $V_L = i_2 (X_L)$

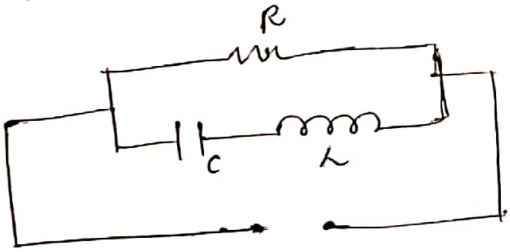
$V_R = i_1 R$

* Here $V_R = V_{Rms}$ &

$|V_L - V_C| = V_{rms}$

Ex:

A box contains L, C & R when 250V DC is applied to the terminals of the box a current of 1A flows in the circuit. when an Ac source of 250V (rms) at 2250 rad/s is connected, a current of 1.25A (rms) flows. It is observed that the current rises with the frequency & becomes max at 4550 rad/sec. Find the values of L, C & R.

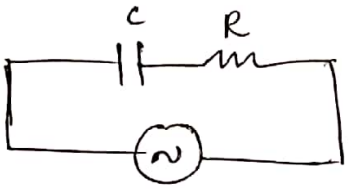


Sol: DC: $E = iR \Rightarrow R = \frac{250}{1} = 250\,\Omega$.

AC: $Z = \frac{250}{1.25} = \frac{250 \times 100}{125} = 200\,\Omega$

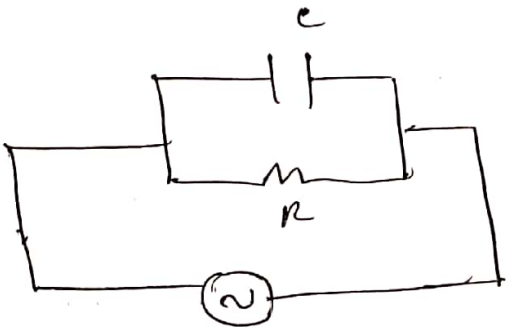
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Ex:



$e = e_0 \sin \omega t$

①



$e = e_0 \sin \omega t$

②

If $Z_1 = 2 Z_2$. Find ω.

Sol:

$$Z_1 = R + X_C$$

$$Z_1 = R - \frac{j}{\omega C}$$

$$\boxed{Z_1 = R - \frac{j}{\omega C}} \quad \text{—①}$$

$$|Z_1| = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

$$\frac{1}{Z_2} = \frac{1}{R} + \frac{1}{X_C}$$

$$\frac{1}{Z_2} = \frac{1}{R} + \frac{1}{\frac{-j}{\omega C}}$$

$$\frac{1}{Z_2} = \frac{1}{R} - \frac{\omega C}{j}$$

$$\boxed{\frac{1}{Z_2} = \frac{1}{R} + j\omega C} \quad \text{—②}$$

$$\left|\frac{1}{Z_2}\right| = \sqrt{\left(\frac{1}{R}\right)^2 + (\omega C)^2}$$

$$\left[R^2 + \left(\frac{1}{\omega C}\right)^2\right]\left[\frac{1}{R^2} + (\omega C)^2\right] = 4$$

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$$R^2\omega^2C^2 + \frac{1}{R^2\omega^2C^2} + 2 = 4$$

$$R^2\omega^2C^2 + \frac{1}{R^2\omega^2C^2} = 2$$

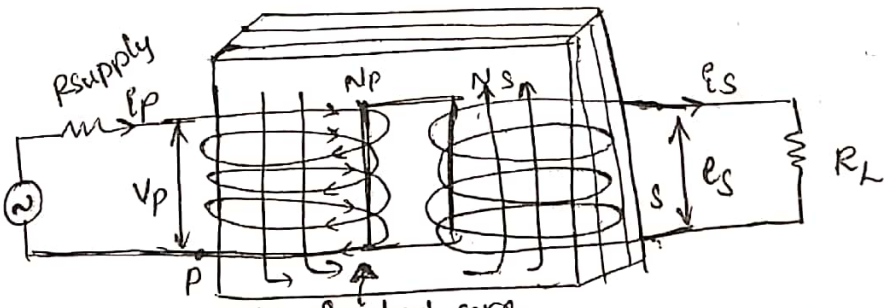
$$\Rightarrow \quad R^2\omega^2C^2 = 1 \quad (\text{or}) \quad \frac{1}{R^2\omega^2C^2} = 1$$

$$\boxed{\omega = \frac{1}{RC}}$$

$$\Rightarrow \boxed{f = \frac{1}{2\pi RC}}$$

Transformers

* It is a device used to transfer electric power from one circuit to another circuit with same frequency by the phenomenon of mutual inductance b/w 2 coils linked with same magnetic flux is called Transformer.
* It works only with AC but not with DC.



$$e_p = i_p R_{supply} + V_p$$

From faraday's law,

$$V_p = N_p \frac{d\phi}{dt} \quad -(1)$$

$$e_s = N_s \cdot \frac{d\phi}{dt} \quad -(2)$$

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* If $R_{supply} = 0 \Rightarrow V_p = e_p$

$$\frac{e_s}{e_p} = \frac{N_s}{N_p}$$ *

* $\frac{N_s}{N_p}$ = Transformer ratio

** Input power given to transformer,

$$P_{in} = e_p i_p$$

& Output power taken from transformer,

$$P_{out} = e_s i_s$$

For ideal transformer,

$$P_{in} = P_{out}$$

$$e_p i_p = e_s i_s$$

$$\frac{e_s}{e_p} = \frac{i_p}{i_s} = \frac{N_s}{N_p}$$

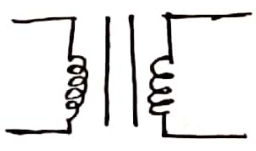
* Effeciency of transformer,

$$\eta = \frac{P_o}{P_i} \times 100\%$$

* In general, $P_i = P_{loss} = P_o$

*** Power loss is due to
- copper loss
- eddy current
- hysteris effect

* Circuit symbol of transformer :-



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Types of Transformers:

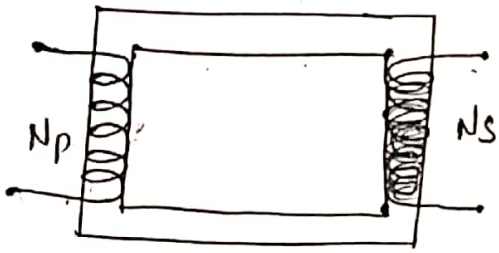
1) Step-up transformer :

$e_s > e_p$

$N_s > N_p$

$i_s < i_p$

electric power transmitts in the form of voltage and current.



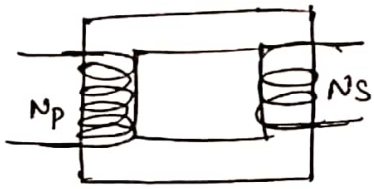
$N_s > N_p$

2) Step-Down transformer :

$e_s < e_p$

$N_s < N_p$

$i_s > i_p$



$N_p > N_s$

eg: bed lamp.

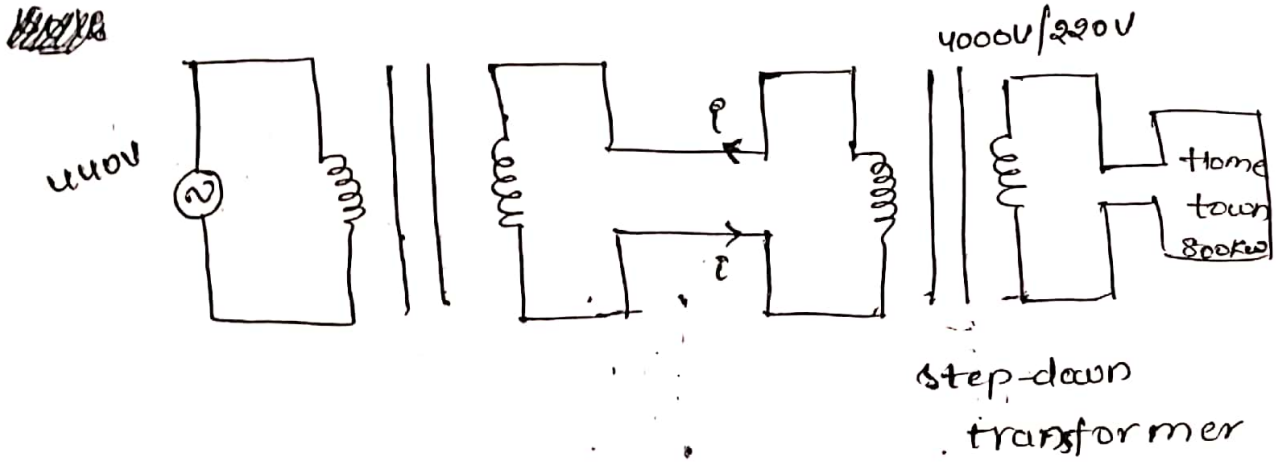
Exe: Hometown demands an electric power of 800kWatt at 220V is located 15km away from the electric plant generating power at 4400V. The resistance of the two line wires caring power is 0.5Ω/km. The town gets power from the lines through a 4000V/220V step down transformer at the substation in the town.

1) Estimate the line power loss in the form of heat.

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2) How much power must the plant supply assuming there is negligible power loss due to linkage.

3) Characterstic the step up transformer.



Sol* current through lines,

$$i = \frac{800 \times 10^3}{4000} = 200\,A$$

1) * Power loss in the wires,

$$P_L = i^2 R$$

$$= (200)^2 (0.5 \times 30)$$

$$\boxed{P_L = 600KW}$$

2) * Power generated at the power plant,

$$P = P_L + P_{Hometown}$$

$$= 600 + 800$$

$$\boxed{P = 1400KW}$$

3) * Voltage across wires of secondary coil of step-up transformer,

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$$V = 4000 + iR$$

$$= 4000 + (200)(0.5 \times 300)$$

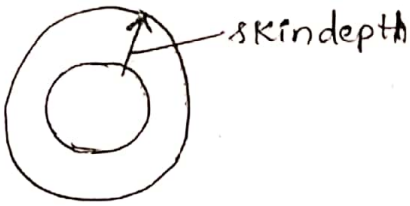
$$\boxed{V = 7000V}$$

* Characteristic ⟶ 440V/7000V step-up transformer.

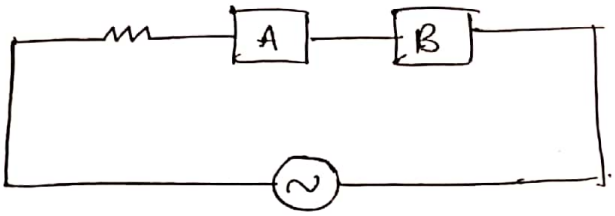
Skin depth:

D.C current flows throughout the volume.





Ex:



4|07|25| Friday

Given $V_R = V\sin\omega t$

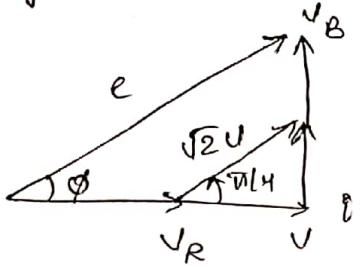
$$V_A = \sqrt{2}V\sin\left(\omega t + \frac{\pi}{4}\right)$$

$$V_B = V\sin\left(\omega t + \frac{\pi}{2}\right)$$

Find the phase diff b/w current & applied voltage.

Sol:

Phasor diagram,



$$\tan\phi = \frac{2V}{2V}$$

$$\boxed{\phi = \frac{\pi}{4}}$$

we have,

$$e = V_R + V_A + V_B$$

$$e = V\sin\omega t + \sqrt{2}V\sin\left(\omega t + \frac{\pi}{4}\right) + V\sin\left(\omega t + \frac{\pi}{2}\right)$$

$$= V\left[\sin(\omega t) + \sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right) + \sin\left(\omega t + \frac{\pi}{2}\right)\right]$$

$$= V\left[\sin\omega t + \sin\omega t + \cos\omega t + \cos\omega t\right]$$

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$= 2V(\sin\omega t + \cos\omega t)$

$$\boxed{e = 2\sqrt{2}V \sin\left(\omega t + \frac{\pi}{4}\right)}$$

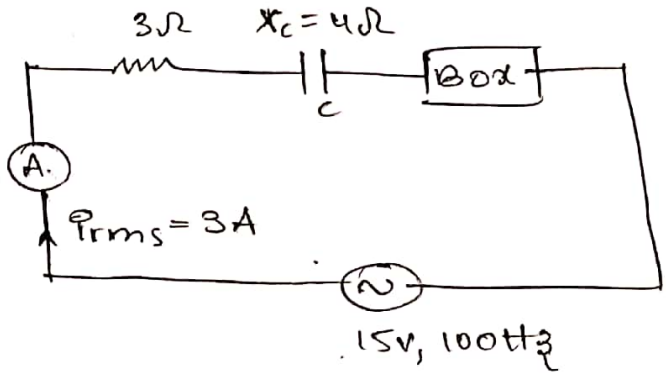
* current eqn,

$$i = i_0 \sin\left(\omega t + \frac{\pi}{4} - \frac{\pi}{4}\right)$$

$$i = i_0 \sin\omega t$$

where $i_0 = \frac{2\sqrt{2}V}{Z}$.

Exg



It is known that the box contains one (or) more elements (R, C (or) L) which elements is/are present inside the box.

with out box,

$$i_{rms} = \frac{V_{rms}}{Z} \Rightarrow \frac{15}{3} = Z \Rightarrow \boxed{Z = 5\Omega}$$

$$V_{RC} = i_{rms}\sqrt{R^2 + X_C^2} = 3\sqrt{9+16} = 15\ V$$

$$= \text{emf of supply}.$$

∴ There is no p.d across box.

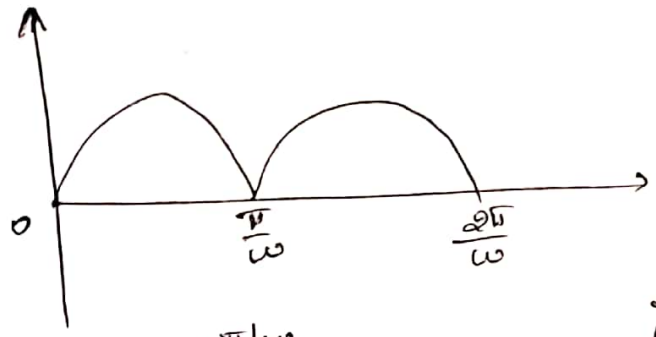
* Box contains L & C in series with $X_L = X_C$.

Exg The avg, rms value of voltage for one cycle is $V = V_0 \sin\omega t$ for $0 < t < \frac{\pi}{\omega}$

$V = -V_0 \sin\omega t$ for $\frac{\pi}{\omega} < t < \frac{2\pi}{\omega}$.

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Sol:



$$V_{avg} = \frac{\int_0^{\pi/\omega} V_0 \sin\omega t \cdot dt + \int_{\pi/\omega}^{2\pi/\omega} -V_0 \sin\omega t \cdot dt}{\int_0^{\frac{2\pi}{\omega}} dt}$$

$$= \frac{\frac{V_0}{\omega}[1-(-1)] + \frac{V_0}{\omega}(1-(-1))}{\frac{2\pi}{\omega}}$$

$$= \frac{4V_0|\phi}{2\pi|\phi} = \frac{2V_0}{\pi}$$

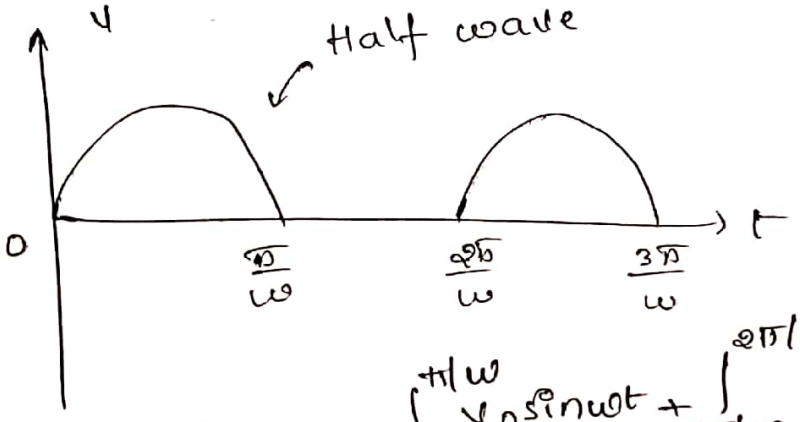
$$V_{rms} = \sqrt{\frac{\omega}{2\pi}\left[\int_0^{\pi/\omega} V_0^2 \sin^2\omega t \cdot dt + \int_{\pi/\omega}^{2\pi/\omega} V_0^2 \sin^2\omega t \cdot dt\right]}$$

$$= \frac{V_0}{\sqrt{2}}$$

Ex: eqn is $V = V_0 \sin\omega t \quad 0 \le t \le \frac{\pi}{\omega}$ Find 1) V_{avg}

$= 0 \quad \frac{\pi}{\omega} \le t \le \frac{2\pi}{\omega}$ 2) V_{rms}

Sol:



$$V_{avg} = \frac{\int_0^{\pi/\omega} V_0 \sin\omega t + \int_{\pi/\omega}^{2\pi/\omega} 0 \cdot dt}{\int_0^{2\pi/\omega} dt}$$

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$$V_{rms} = \sqrt{\frac{\int_0^{\pi/\omega} V_0^2 \sin^2 \omega t \cdot dt + \int_{\pi/\omega}^{2\pi/\omega} 0 \cdot dt}{\int_0^{2\pi} dt}}$$

Ex: An AC source producing emf $V = V_0(\sin\omega t + \sin 2\omega t)$ is connecting in series with a capacitor & a resistor the current found in the circuit is $i = i_1 \sin(\omega t + \phi_1) + i_2 \sin(2\omega t + \phi_2)$ then 1) $i_1 = i_2$

2) $i_1 < i_2$

3) $i_1 > i_2$

4) i_1 may be less than or equal or greater than i_2.

Sol: $i_1 = \frac{V_0}{Z_1}$; $i_2 = \frac{V_0}{Z_2}$ $\quad Z_2 = \sqrt{R^2 + \left(\frac{1}{2\omega C}\right)^2}$

$Z_1 = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$ clearly $Z_1 > Z_2$

$\boxed{i_1 < i_2}$

Ex: A dc ammeter and an AC thermal ammeter are connected to a circuit in series when a dc is passed through the circuit the dc ammeter shows $i_1 = 6$ A when a sinusoidal Alternating current flows

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through the circuit, the ac ammeter shows $i_2 = 8A$. If both i_1 & i_2 are passed in the circuit what are the readings of meters.

Sol: when both are ~~place~~ passed

DC ammeter reading is 6A.

AC ammeter reading is rms value of total current.

$$i = 6 + 8\sqrt{2}\sin\omega t$$

DC ammeter

$$i_{avg} = \frac{\int_0^T (6 + 8\sqrt{2}\sin\omega t)\cdot dt}{T}$$

$$= 6\ A$$

AC ammeter

$$i_{rms} = \sqrt{\frac{1}{T}\int_0^T \left(6 + 8\sqrt{2}\sin\omega t\right)^2 \cdot dt}$$

$$= 10\ A$$

Ex: Power factor of an L-R series circuit is 0.6 & that of a C-R series circuit is 0.5. If the elements L, C & R of the 2 circuits are joined in series the power factor of the circuit is found to be "one". Find the ratio of the resistance in the L-R circuit to the resistance in

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C-R circuit

Sol^n

$$\frac{R_1}{\sqrt{R_1^2 + X_L^2}} = \frac{3}{5}$$

$$\frac{25}{9} R_1^2 = R_1^2 + X_L^2$$

$$R_1^2 = \frac{9}{16} X_L^2$$

$$\boxed{R_1 = \frac{3}{4} X_L}$$

$$\frac{R_2}{\sqrt{R_2^2 + X_C^2}} = \frac{1}{2}$$

$$R_2 = \frac{X_C}{\sqrt{3}}$$

now, $$\cos\phi = \frac{R_1 + R_2}{\sqrt{(R_1 + R_2)^2 + (X_L - X_C)^2}} = 1$$

$$X_L - X_C = 0$$

$$X_L = X_C$$

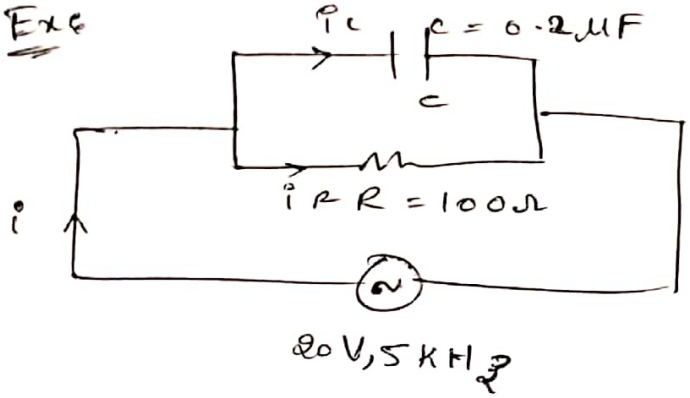
$$\frac{R_1}{R_2} = \frac{3/4\, X_L}{\frac{X_C}{\sqrt{3}}}$$

$$= \frac{3}{4} \times \sqrt{3} \times \frac{X_L}{X_C}$$

$$\boxed{\frac{R_1}{R_2} = \frac{3\sqrt{3}}{4}}$$

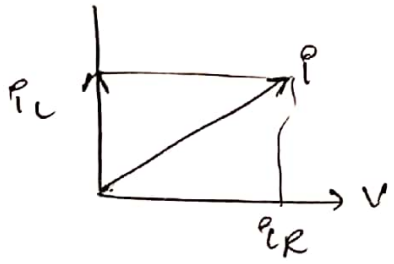
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Exe



Find i_c, i_R, i.

Sol



$$\tan\phi = \frac{i_c}{i_R}$$

$$\frac{1}{Z} = \frac{1}{R} + j\omega C$$

$$\left|\frac{1}{Z}\right| = \sqrt{\left(\frac{1}{R}\right)^2 + \omega^2 C^2}$$

$$i_{rms} = \frac{V_{rms}}{Z}$$

$$= 20\sqrt{\left(\frac{1}{100}\right)^2 + 4\times10^{-14}\times100\times\pi^2\times10^6}$$

$$= 20\sqrt{10^{-4} + 4\times10^{-6}\times\pi^2}$$

$$=$$

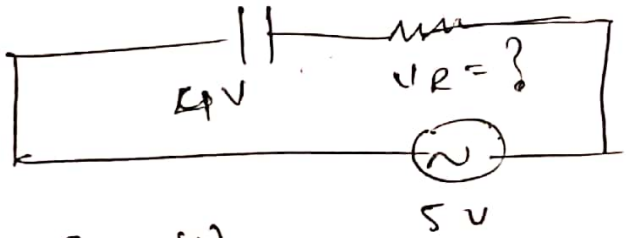
$$i_R = \frac{20}{100} = \frac{1}{5}$$

$$i_c = \frac{20}{\frac{1}{\omega C}}$$

$$=$$

$$i = \sqrt{i_c^2 + i_R^2}$$

Exe



At any time 't'

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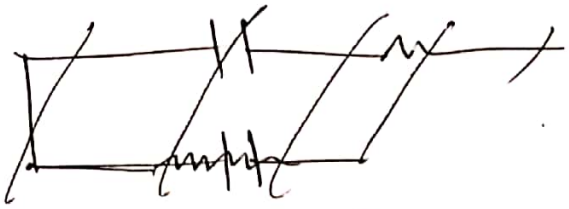
$$e = V_C + V_R$$

$$5 = \pm 4 \pm V_R$$

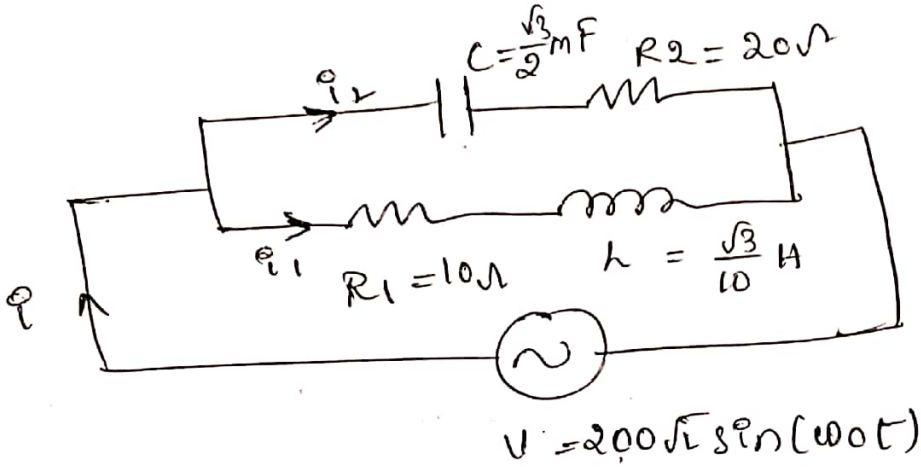
$$\pm V_R = 5 + 4 \text{ (or) } 5 - 4$$

$$\boxed{V_R = 9 \text{ (or) } 1 V}$$

Ex 6



Ex 6



Find 1) phase diff b/w i_1 & i_2 ____90°____

2) At some instant current in LR_1 is ____ 10A At same instant current in CR_2 is ____ ,

3) At some instant current (i_1) in the circuit is $10\sqrt{2}$A then at the same instant $i =$ ____

Sol 6 for L-R

$$i_{0_1} = \frac{200\sqrt{2}}{\sqrt{100 + \left(10\phi \times \frac{\sqrt{3}}{10}\right)^2}} = \frac{200\sqrt{2}}{\sqrt{400}}$$

$$= 10\sqrt{2}$$

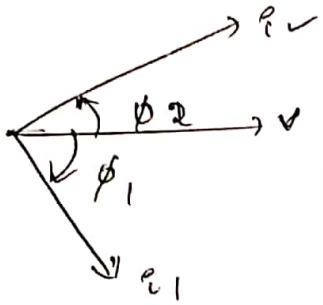
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for C-R

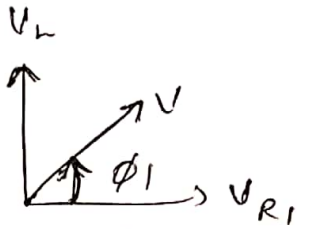
$$i_{0_2} = \frac{200\sqrt{2}}{\sqrt{400 + \left(\frac{1}{100} \times \frac{22}{\sqrt{3}} \times 10^{+3}\right)^2}}$$

$$= \frac{200\sqrt{2}}{\sqrt{400 + \frac{400}{3}}}$$

$$= 5\sqrt{6}$$

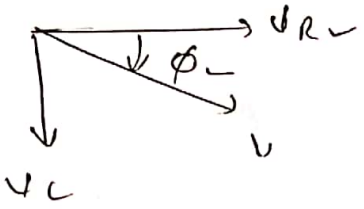


$\phi_1 + \phi_2 = 90°$



$$\tan\phi_1 = \frac{V_L}{V_{R_1}}$$

In this case if 2 balls are connected in 2 branches the bulb glows alternatively.



$$\tan\phi_2 = \frac{V_C}{V_{R_2}}$$

eqn of currents,

$$i_1 = i_{0_1} \sin\left(\omega t - \frac{\pi}{3}\right) \qquad i_{0_1} = 10\sqrt{2}$$

$$i_2 = i_{0_2} \sin\left(\omega t + \frac{\pi}{6}\right) \qquad i_{0_2} = 5\sqrt{6}$$

2) given, $i_1 = 10A$

$$10 = 10\sqrt{2} \sin(100t - \pi/3)$$

$$100t = \frac{7\pi}{12}$$

$$i_2 = 5\sqrt{6} \sin\left(\frac{7\pi}{12} + \frac{\pi}{6}\right) = 5\sqrt{3}$$

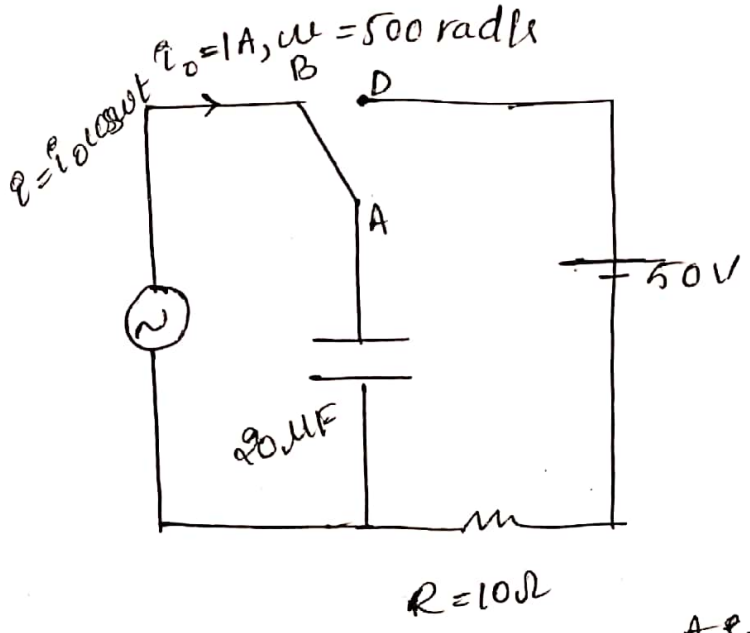
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3) $10\sqrt{1} = 10\sqrt{1}\sin\left(100t - \frac{\pi}{3}\right)$

$$100t = \frac{5\pi}{6}$$

$$i_2 = 5\sqrt{6} \times \sin\pi = 0 .$$

Ex 6



At $t = \frac{7\pi}{6\omega}$, the key is switched from B to D. Now onwards only A & D are connected

a total charge 'Q' flows from battery to charge the capacitor fully. Verify the correct statements.

a) Mag of max charge on capacitor before $t = \frac{7\pi}{6\omega}$ is 1×10^{-3} C

b) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$ is

c) Immediately after A is connected to D, the current in R is 10A.

d) $Q = 2 \times 10^{-3}$ C.

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Sol6

$$\text{at} \quad t = \frac{7\pi}{6\omega}$$

$$i = -\frac{\sqrt{3}}{2} A$$

$$\Rightarrow q = \int i_0 \cos\omega t \cdot dt$$

$$= \frac{i_0}{\omega} \sin\omega t$$

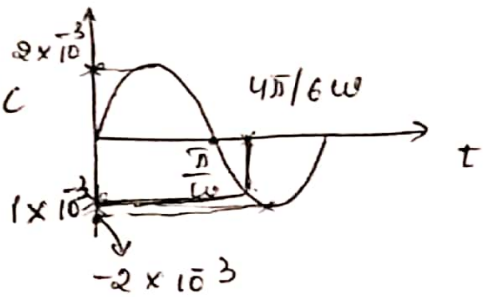
$$\text{at } t = \frac{7\pi}{6\omega}$$

$$\Rightarrow |q| = \frac{i_0}{\omega} \sin\frac{7\pi}{6}$$

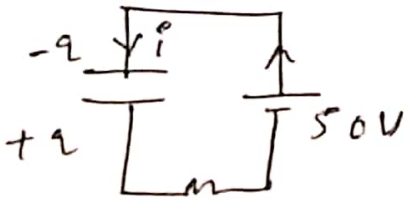
$$= \frac{1}{500} \times \frac{1}{2}$$

$$\boxed{q = 10^{-3} C}$$

$\frac{i_0}{\omega}$ = max charge = $2 \times 10^{-3} C$



Immediately when A & D are connected



$$50 + \frac{q}{C} - iR = 0$$

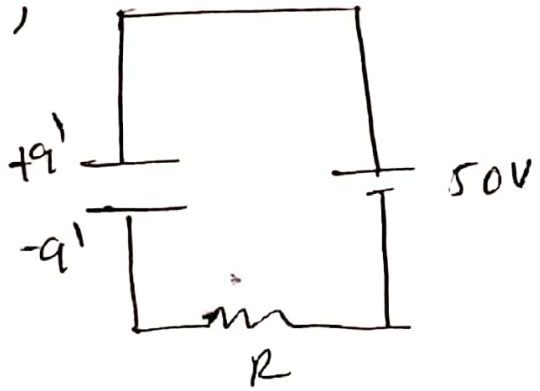
$$i = \frac{50 + q/C}{R}$$

$$= \frac{50 + \frac{10^{-3}}{20\times10^{-6}}}{10}$$

$$\boxed{i = 10A}$$

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Finally,



$q' = 20 \times 10^{-6} \times 50$

$q' = 10^{-3}\ C$

$\Rightarrow \boxed{Q = 2 \times 10^{-3}\ C}$

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